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(54) DRILLING FRAMEWORK

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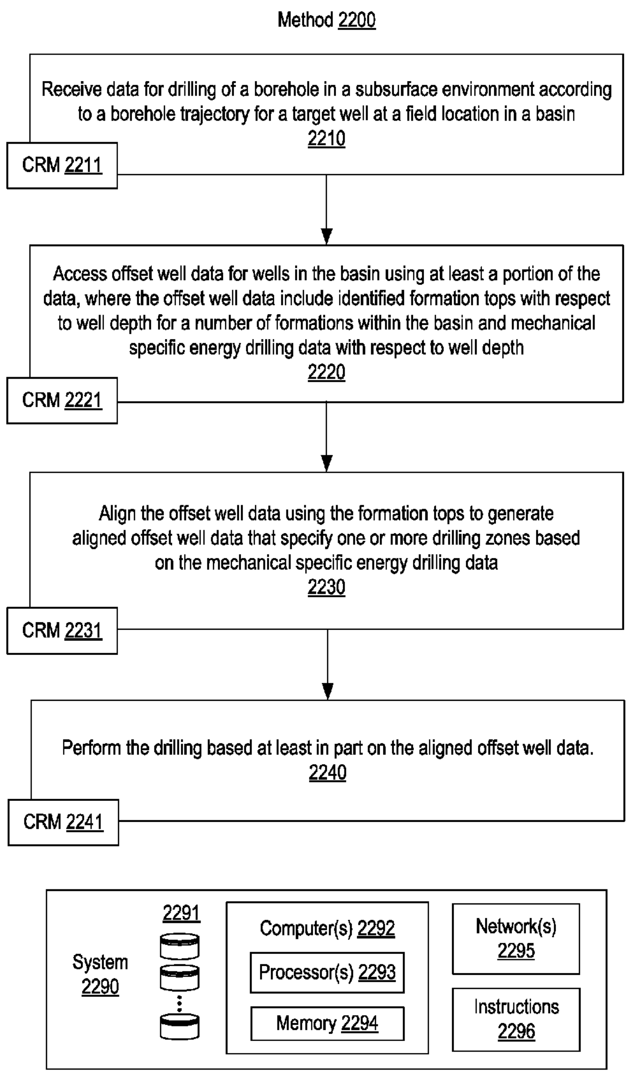
(57) **ABSTRACT**

A method may include receiving data for drilling of a borehole in a subsurface environment according to a borehole trajectory for a target well at a field location in a basin; accessing offset well data for wells in the basin using at least a portion of the data, where the offset well data include identified formation tops with respect to well depth for a number of formations within the basin and mechanical specific energy drilling data with respect to well depth; aligning the offset well data using the formation tops to generate aligned offset well data that specify one or more drilling zones based on the mechanical specific energy drilling data; and performing the drilling based at least in part on the aligned offset well data.



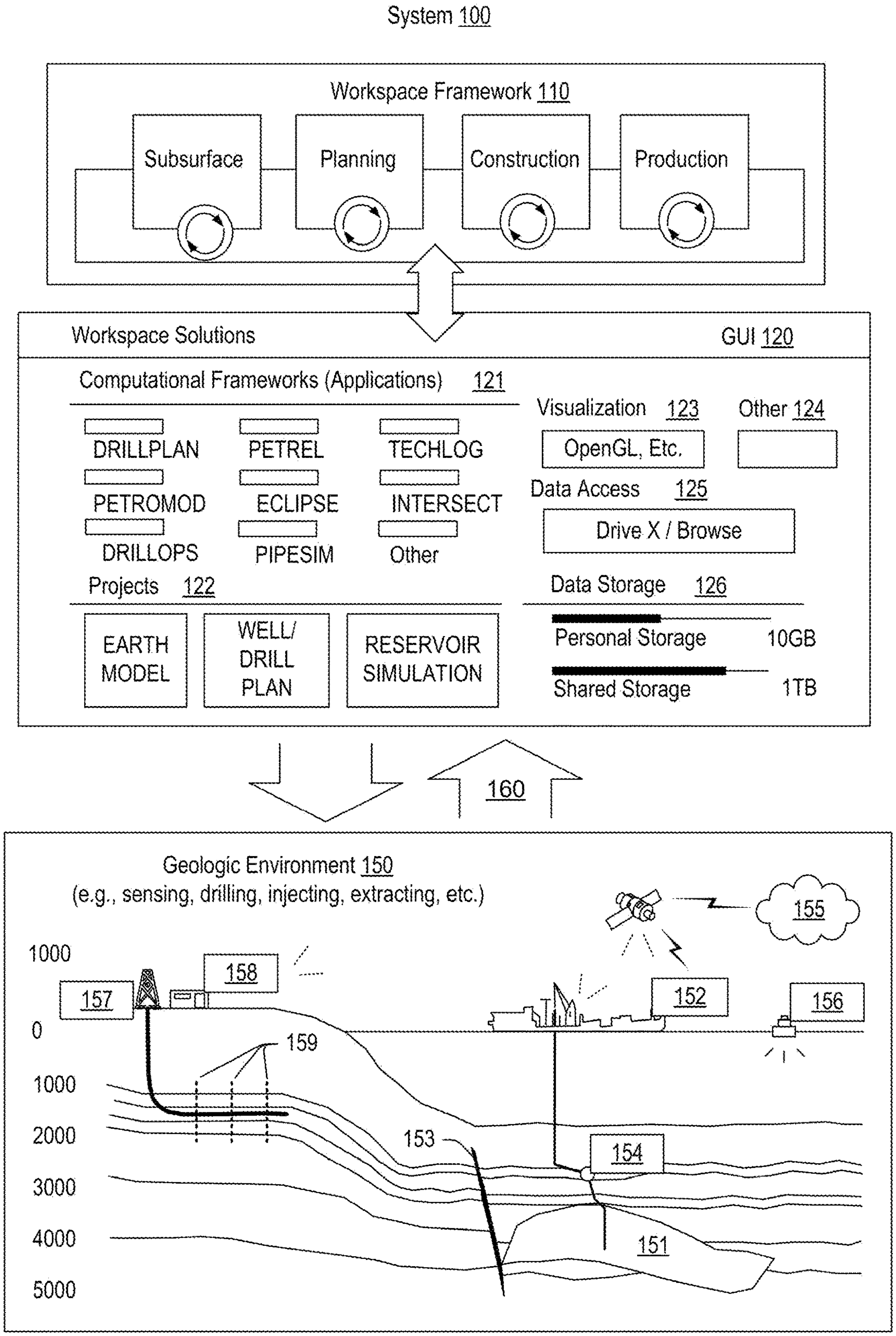
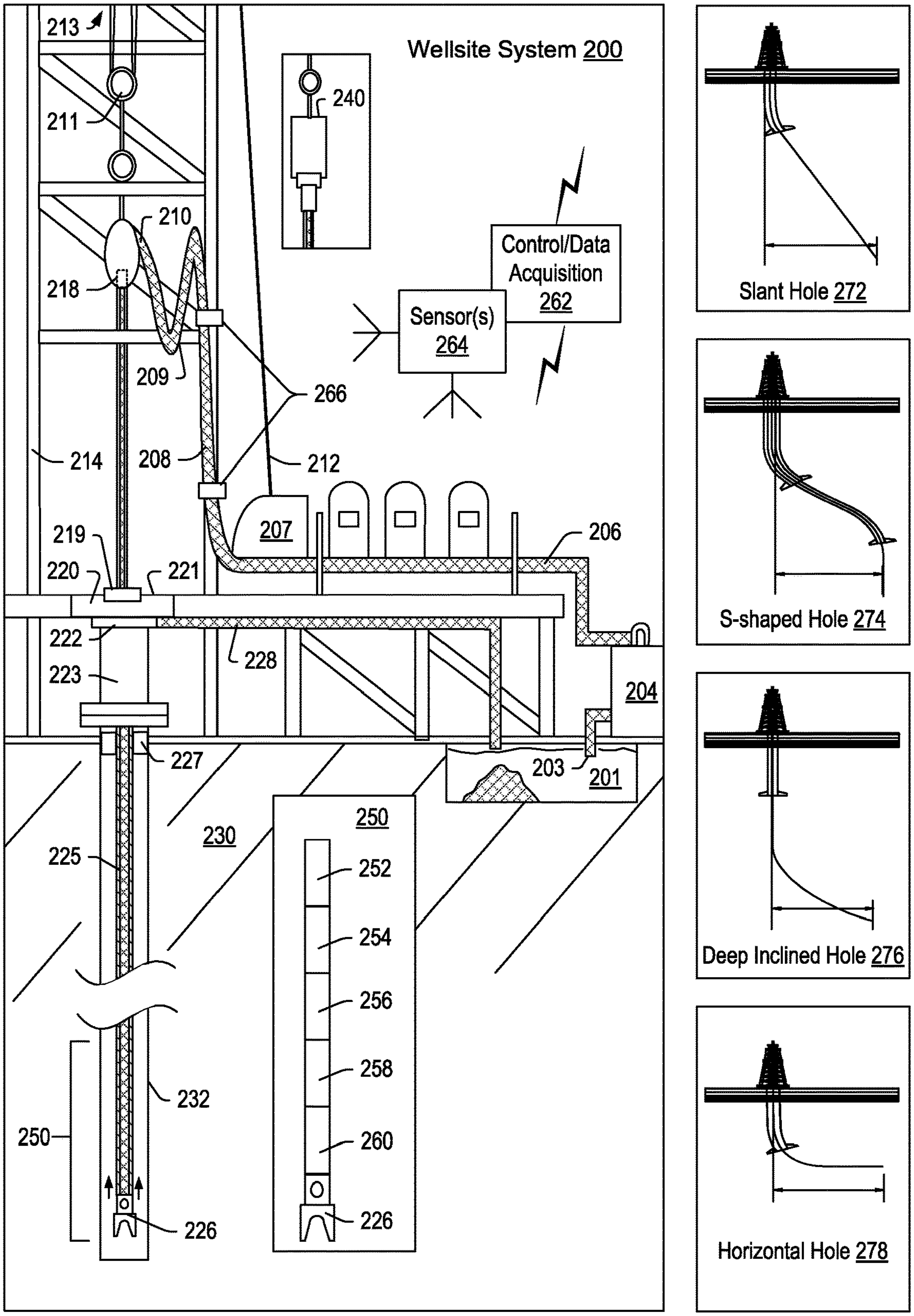

System 100
Workspace Framework 110
Subsurface
Planning
Construction
Production
Workspace Solutions
GUI 120
Computational Frameworks (Applications) 121
DRILLPLAN
PETREL
TECHLOG
PETROMOD
ECLIPSE
INTERSECT
DRILLOPS
PIPESIM
Other
Visualization 123
OpenGL, Etc.
Other 124
Data Access 125
Drive X / Browse
Projects 122
EARTH MODEL
WELL/ DRILL PLAN
RESERVOIR SIMULATION
Data Storage 126
Personal Storage
10GB
Shared Storage
1TB
160
Geologic Environment 150
(e.g., sensing, drilling, injecting, extracting, etc.)
155
1000
158
157
152
156
0
159
1000
2000
153
154
3000
4000
151
5000

Fig. 1



Wellsite System 200
213
211
240
210
218
209
Control/Data Acquisition 262
Sensor(s) 264
266
214
208
212
207
206
219
220
221
222
228
223
204
227
203
201
230
250
225
252
254
256
258
232
260
250
226
226
Slant Hole 272
S-shaped Hole 274
Deep Inclined Hole 276
Horizontal Hole 278

Fig. 2

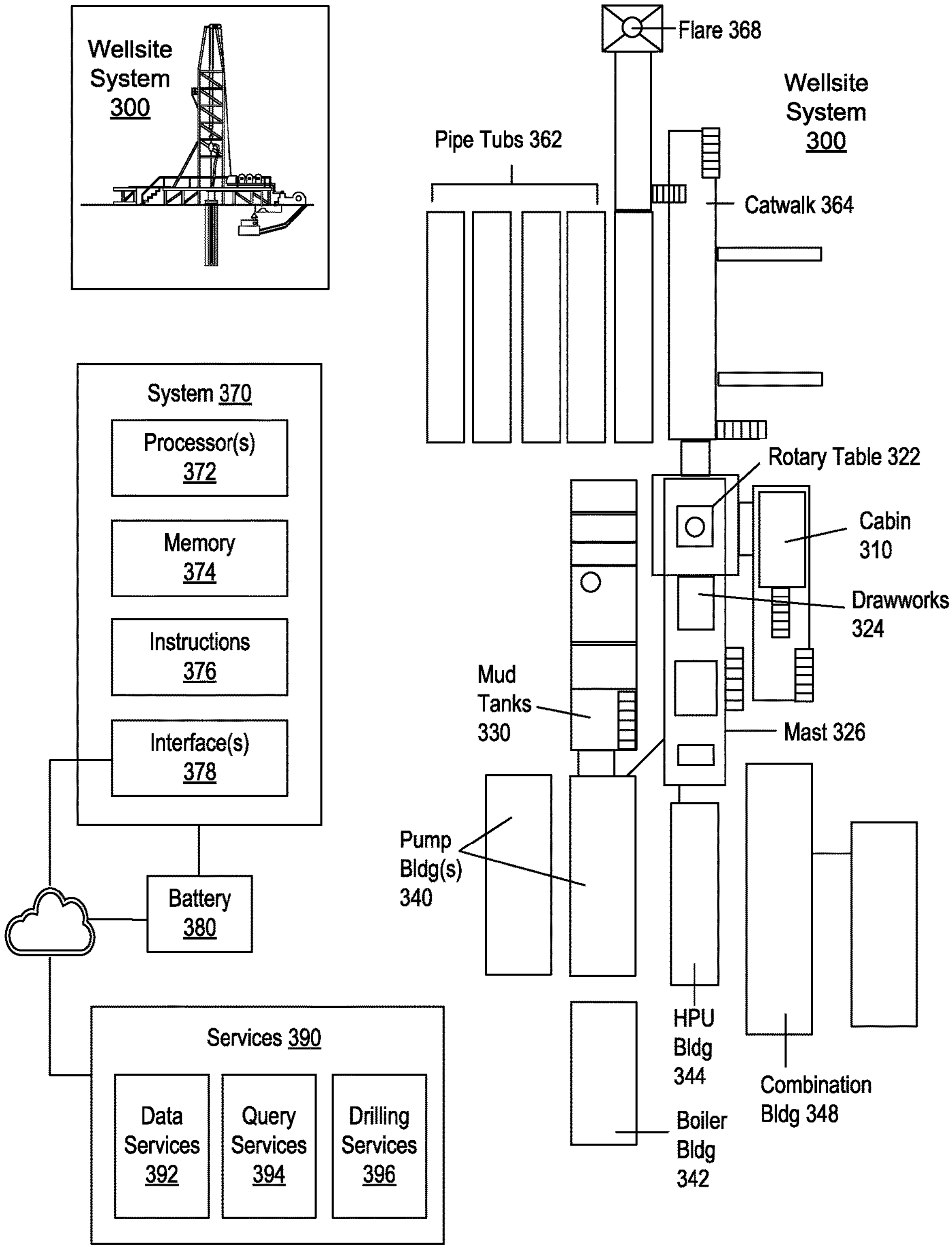

Wellsite System 300
System 370
Processor(s) 372
Memory 374
Instructions 376
Interface(s) 378
Battery 380
Services 390
Data Services 392
Query Services 394
Drilling Services 396
Flare 368
Wellsite System 300
Pipe Tubs 362
Catwalk 364
Rotary Table 322
Cabin 310
Drawworks 324
Mud Tanks 330
Mast 326
Pump Bldg(s) 340
HPU Bldg 344
Combination Bldg 348
Boiler Bldg 342

Fig. 3

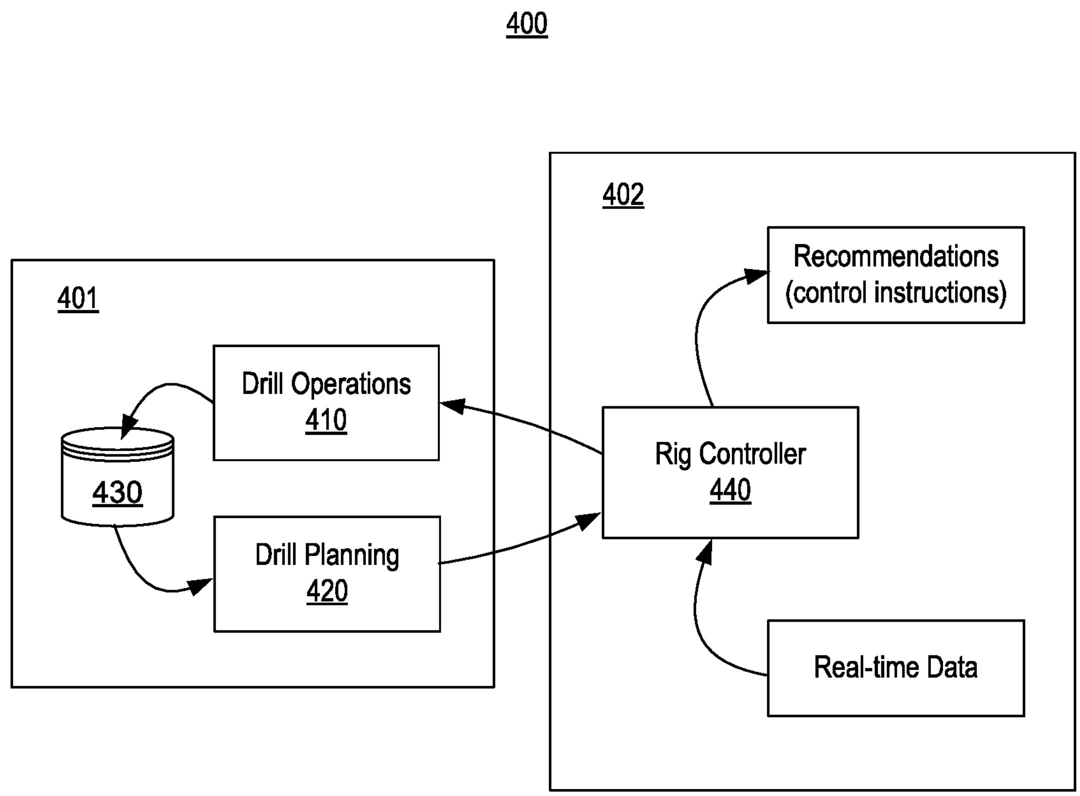

400
402
Recommendations (control instructions)
401
Drill Operations 410
430
Rig Controller 440
Drill Planning 420
Real-time Data

Fig. 4

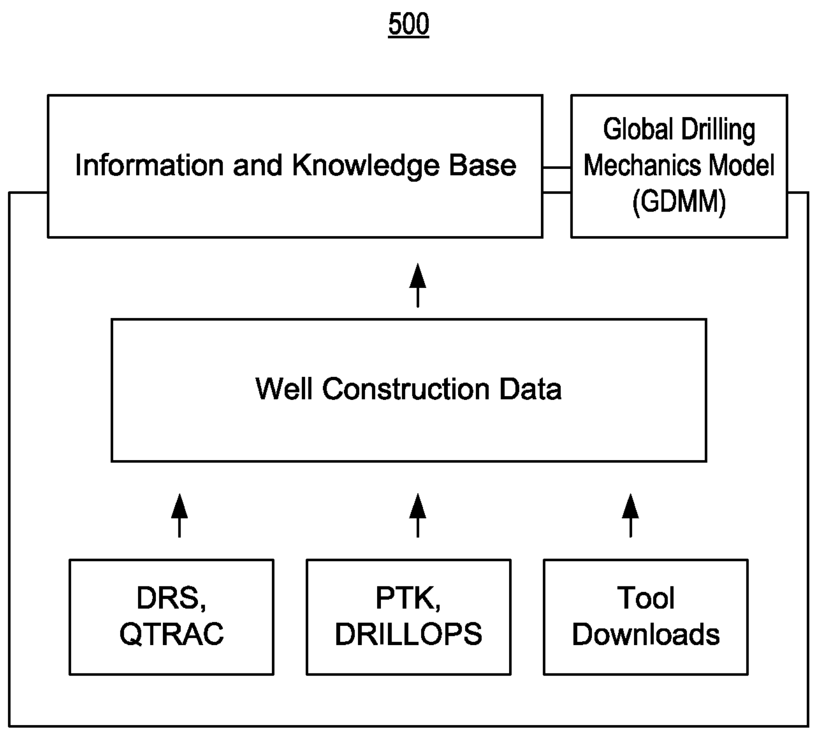

500
Information and Knowledge Base
Global Drilling Mechanics Model (GDMM)
Well Construction Data
DRS, QTRAC
PTK, DRILLOPS
Tool Downloads

Fig. 5

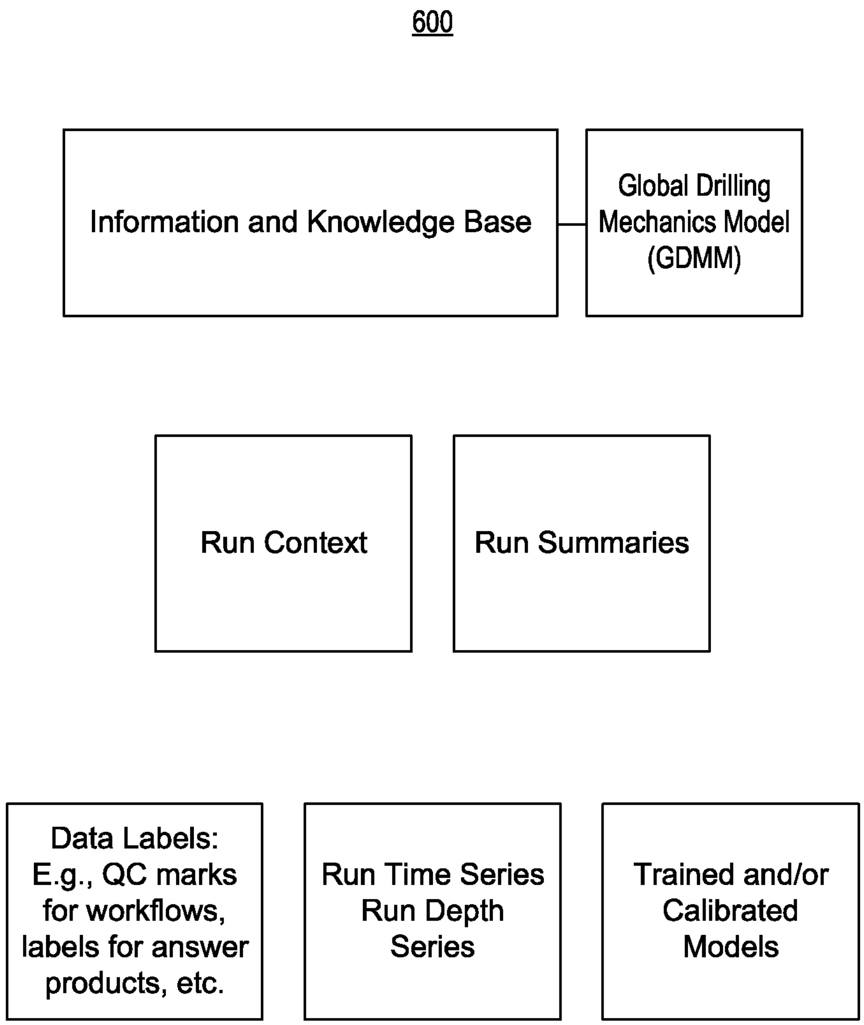

600
Information and Knowledge Base
Global Drilling Mechanics Model (GDMM)
Run Context
Run Summaries
Data Labels: E.g., QC marks for workflows, labels for answer products, etc.
Run Time Series Run Depth Series
Trained and/or Calibrated Models

Fig. 6

700

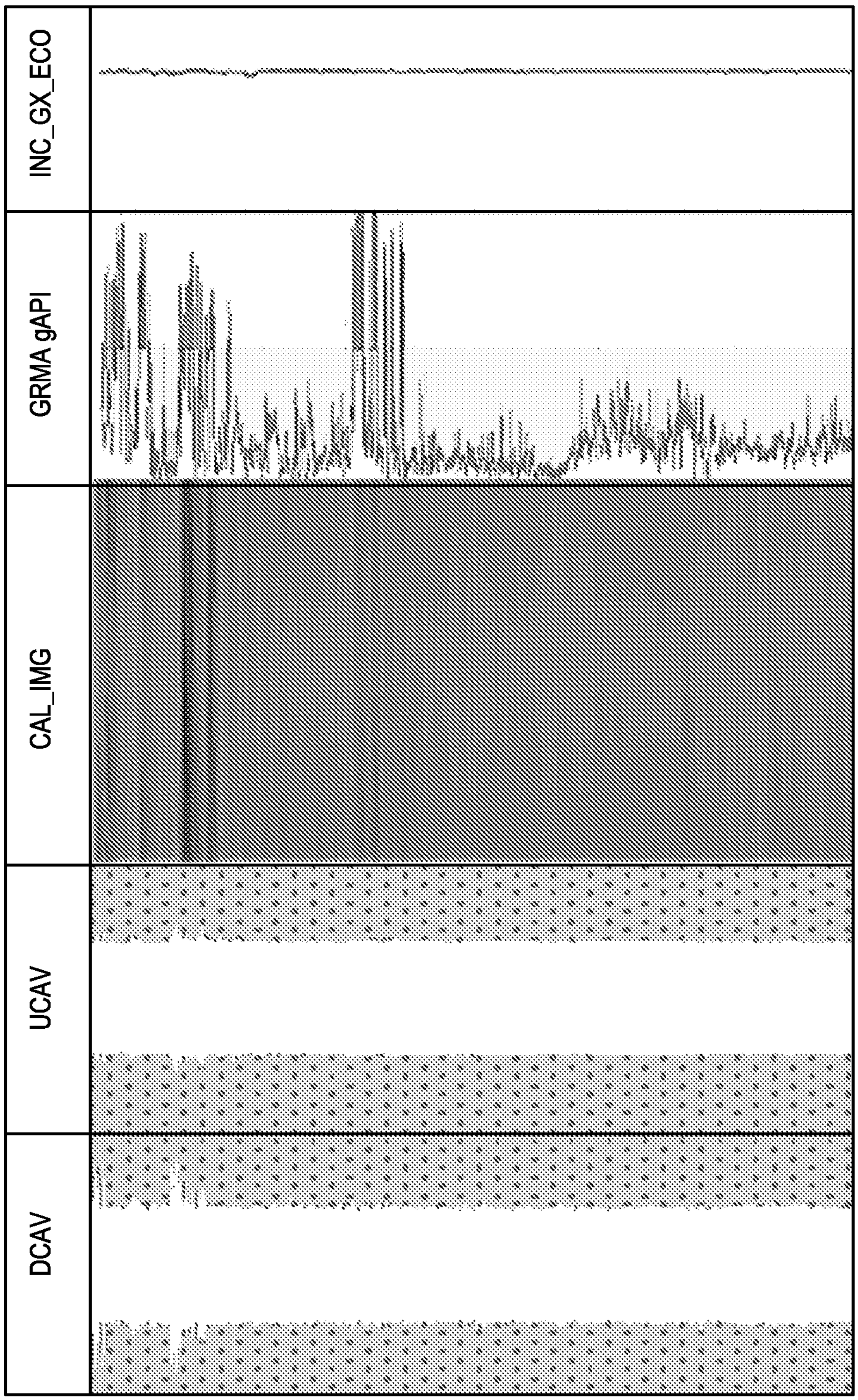
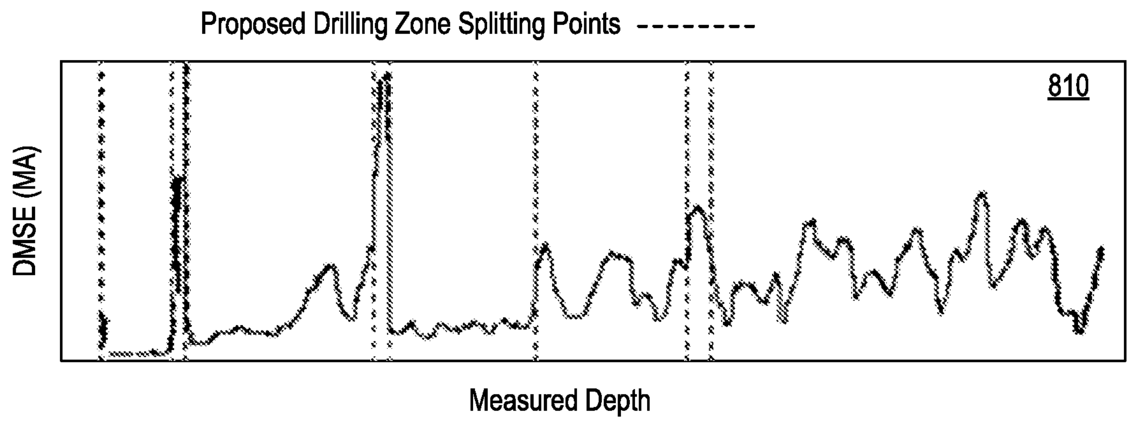


Fig. 7

800


Proposed Drilling Zone Splitting Points
810
DMSE (MA)
Measured Depth

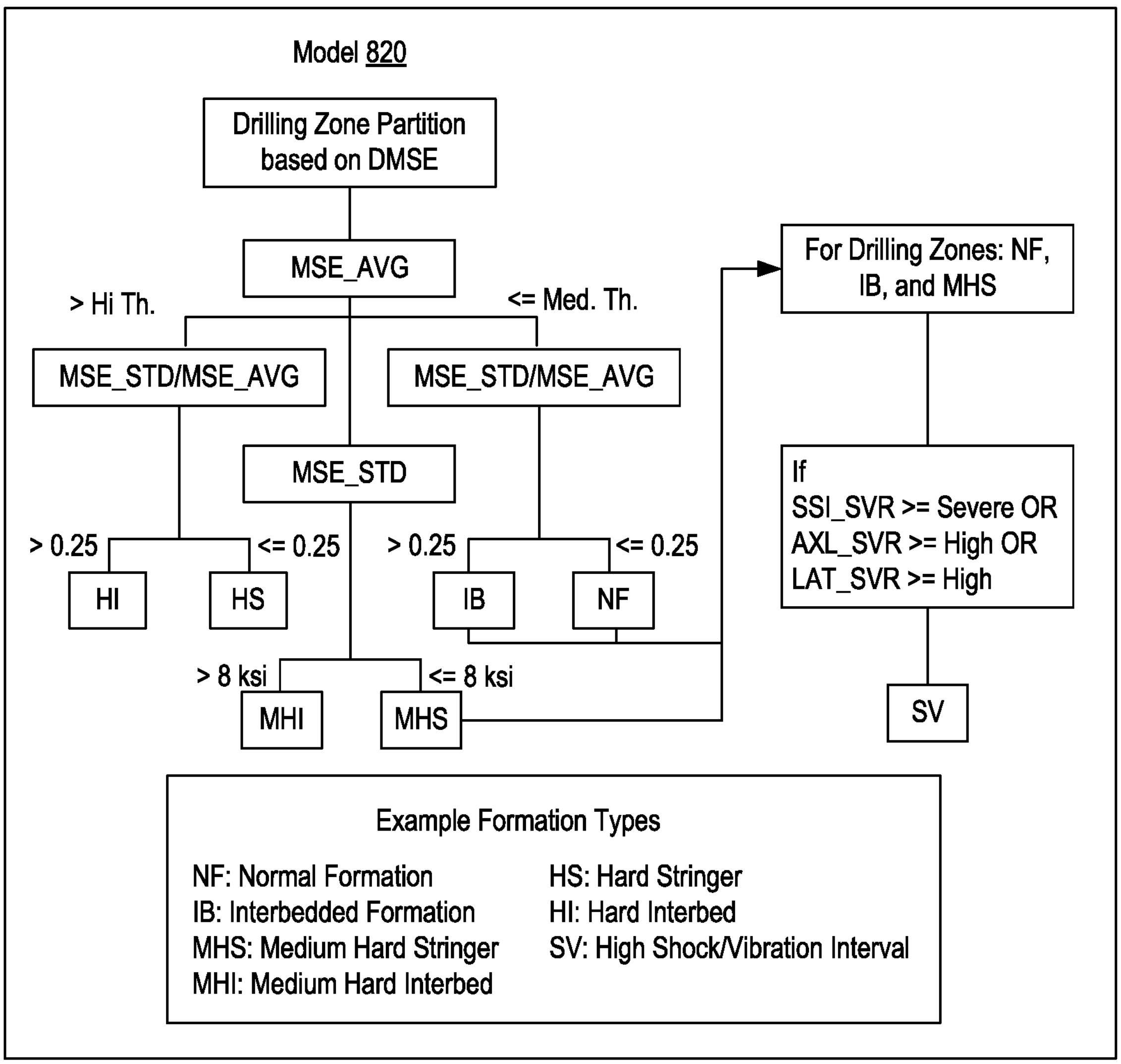

Model 820
Drilling Zone Partition based on DMSE
MSE_AVG
> Hi Th.
<= Med. Th.
MSE_STD/MSE_AVG
MSE_STD/MSE_AVG
MSE_STD
> 0.25
<= 0.25
> 0.25
<= 0.25
HI
HS
IB
NF
> 8 ksi
<= 8 ksi
MHI
MHS
For Drilling Zones: NF, IB, and MHS
If
SSI_SVR >= Severe OR
AXL_SVR >= High OR
LAT_SVR >= High
SV
Example Formation Types
NF: Normal Formation
IB: Interbedded Formation
MHS: Medium Hard Stringer
MHI: Medium Hard Interbed
HS: Hard Stringer
HI: Hard Interbed
SV: High Shock/Vibration Interval

Fig. 8

900

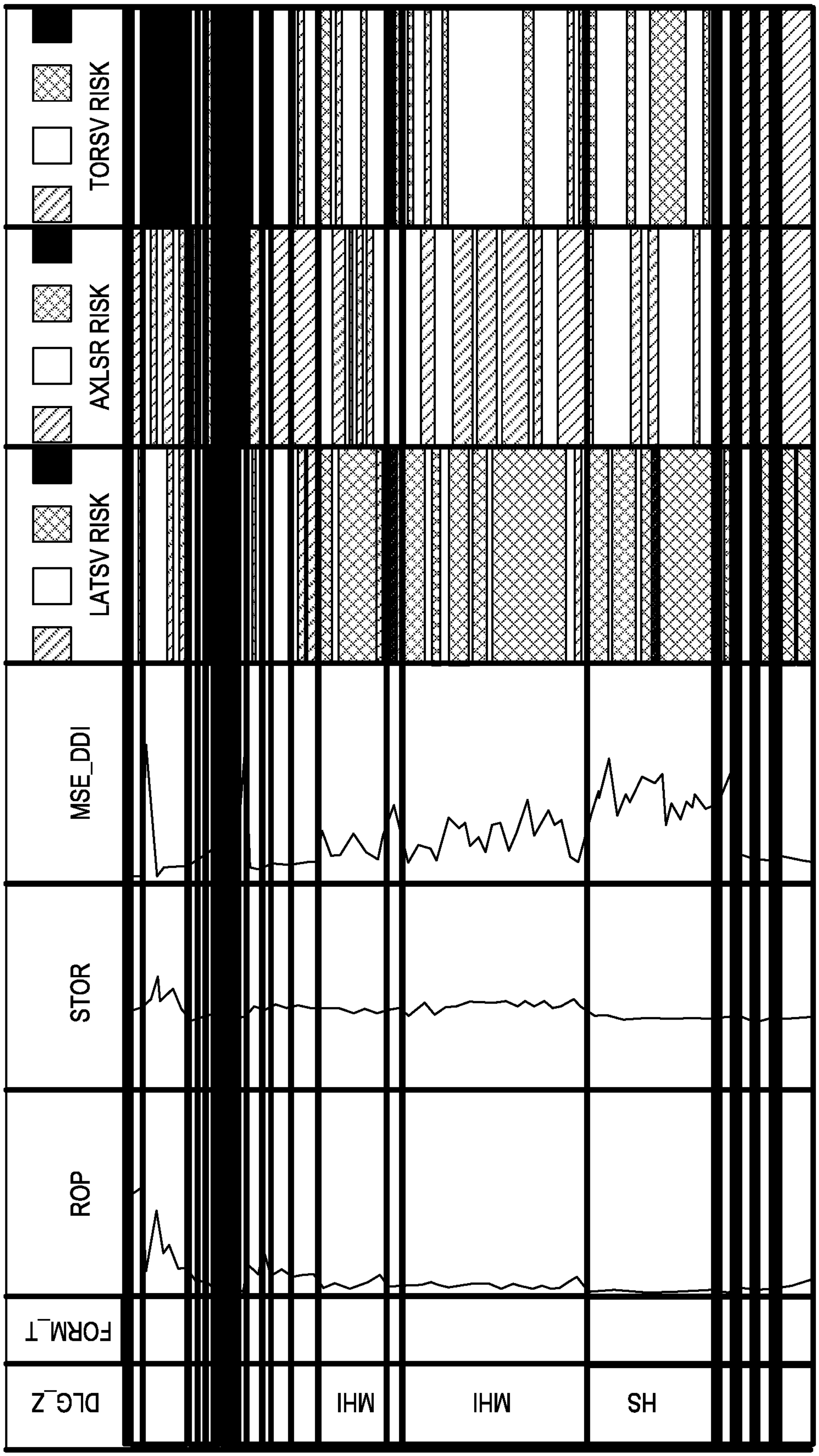

TORSV RISK
AXLSR RISK
LATSV RISK
MSE_DDI
STOR
ROP
FORM_T
DLG_Z
MHI
MHI
HS

Fig. 9

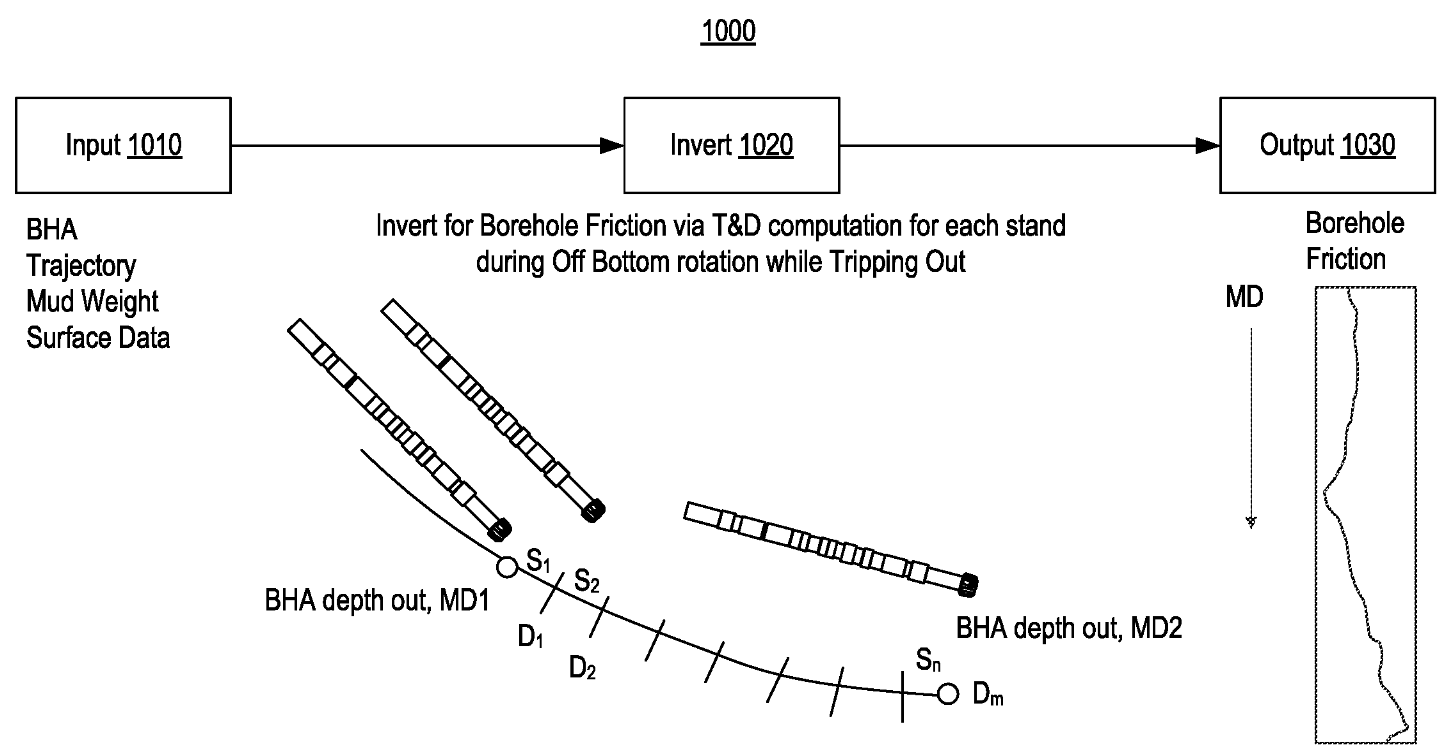
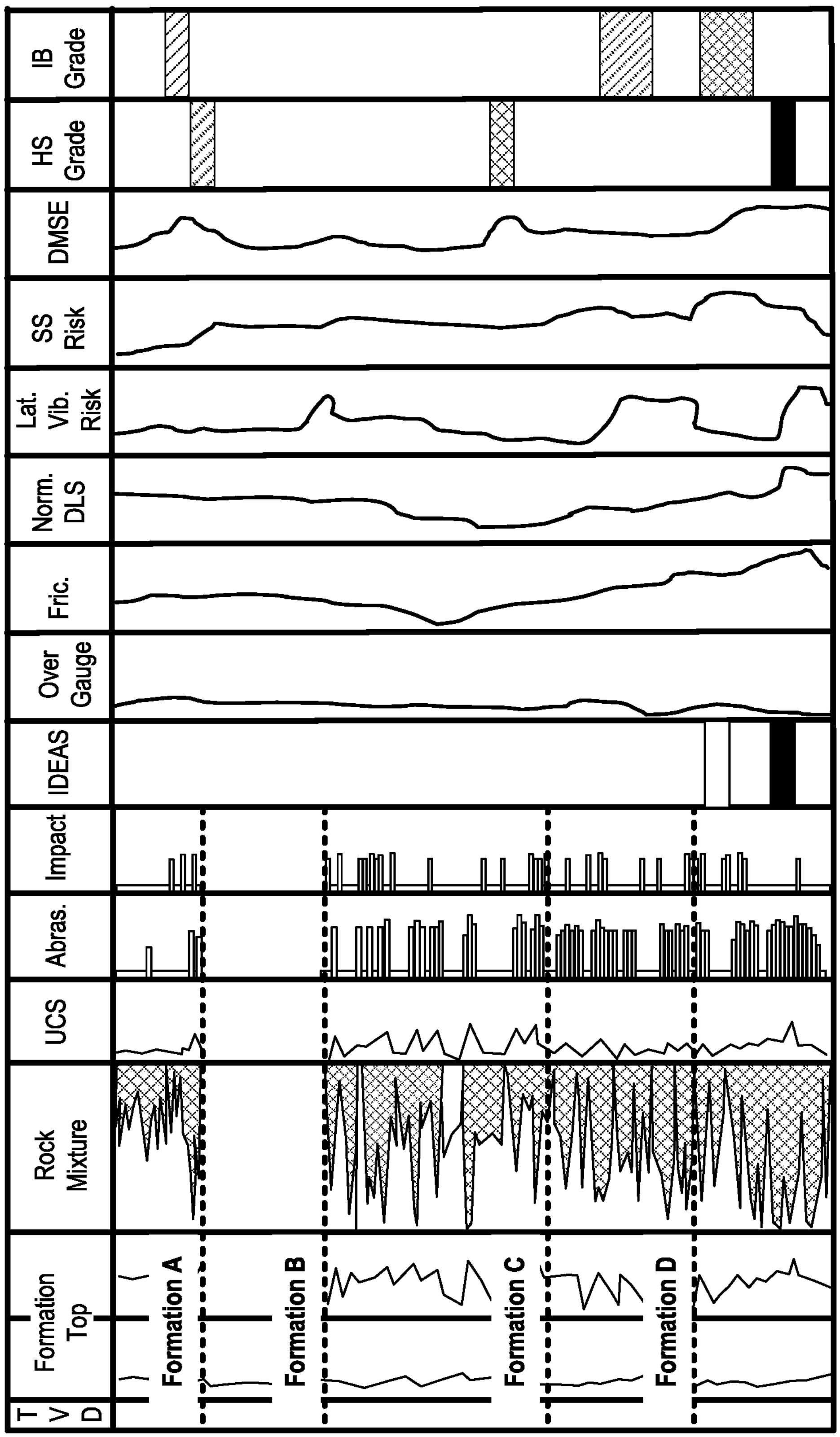

1000
Input 1010
Invert 1020
Output 1030
BHA
Trajectory
Mud Weight
Surface Data
Invert for Borehole Friction via T&D computation for each stand
during Off Bottom rotation while Tripping Out
Borehole
Friction
MD
BHA depth out, MD1
BHA depth out, MD2
S_1
S_2
S_n
D_1
D_2
D_m

Fig. 10

1100



IB Grade
HS Grade
DMSE
SS Risk
Lat. Vib. Risk
Norm. DLS
Fric.
Over Gauge
IDEAS
Impact
Abras.
UCS
Rock Mixture
Formation Top
Formation A
Formation B
Formation C
Formation D
T V D

Fig. 11

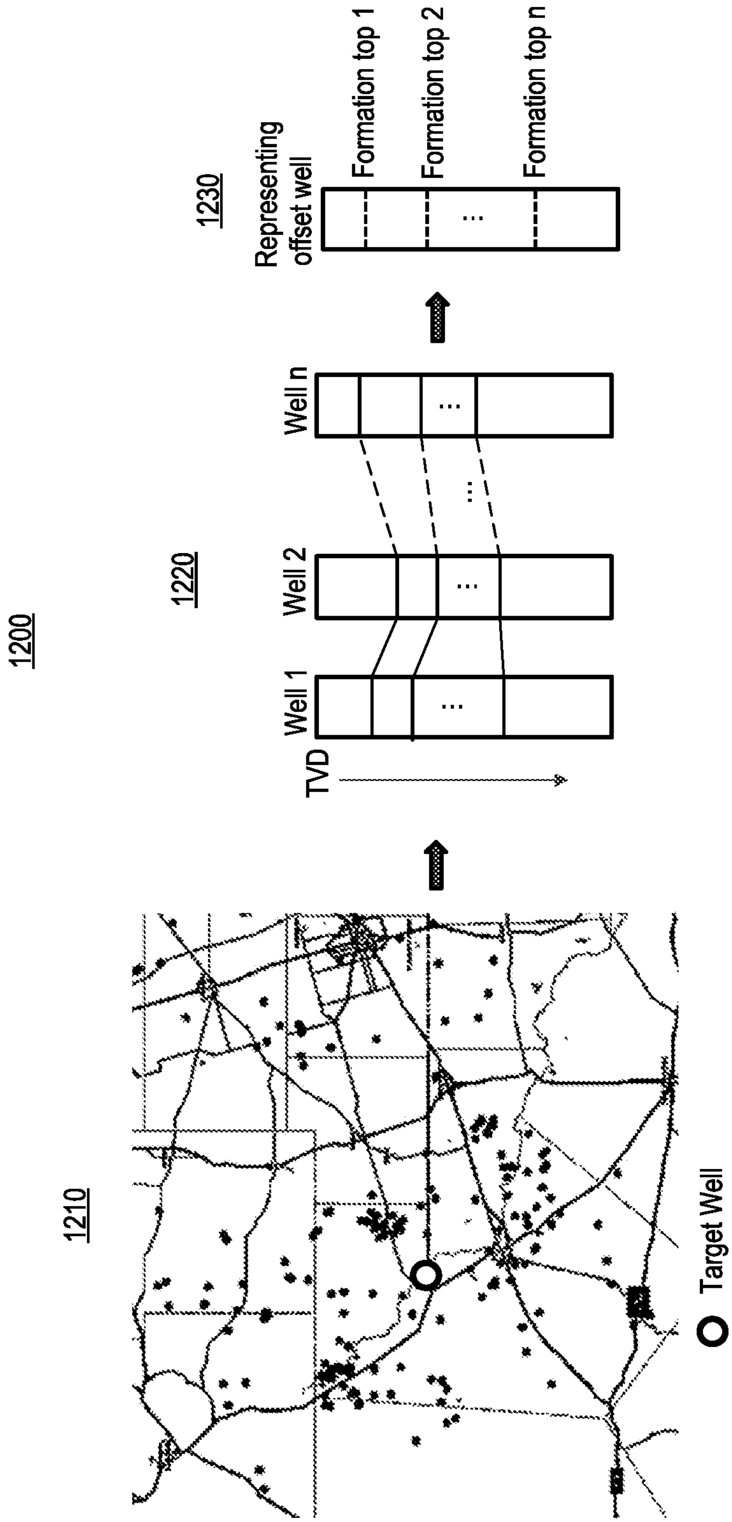

1200
1210
Target Well
1220
TVD
Well 1
Well 2
Well n
...
1230
Representing offset well
Formation top 1
Formation top 2
Formation top n

Fig. 12

1300

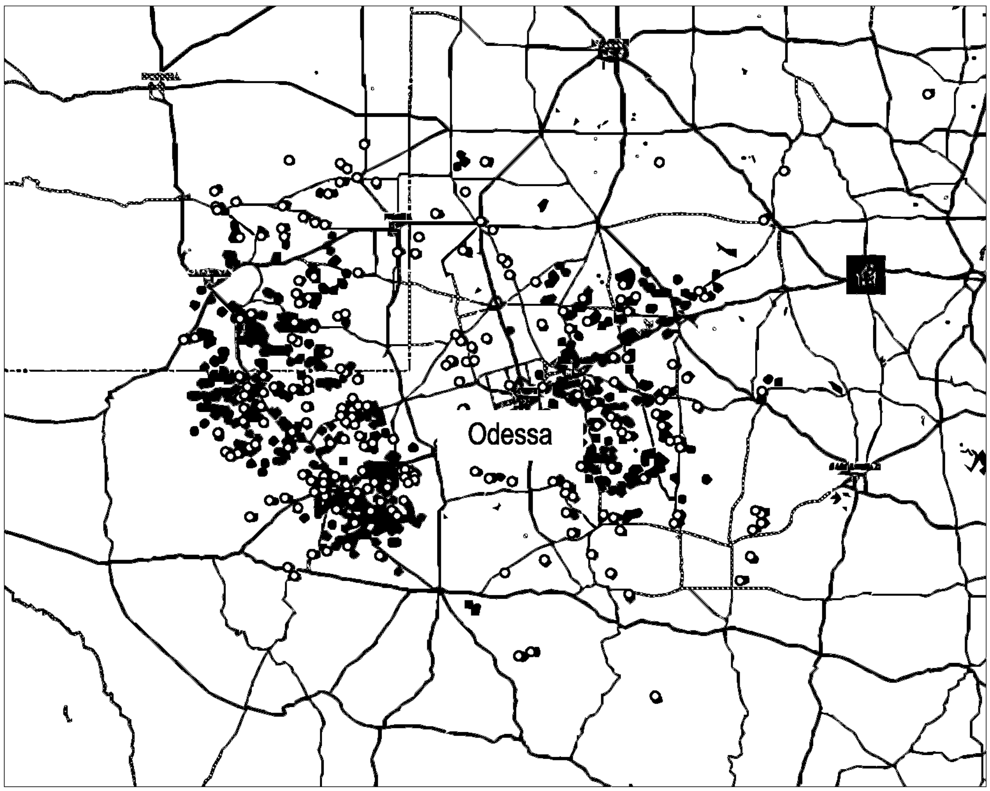


Fig. 13

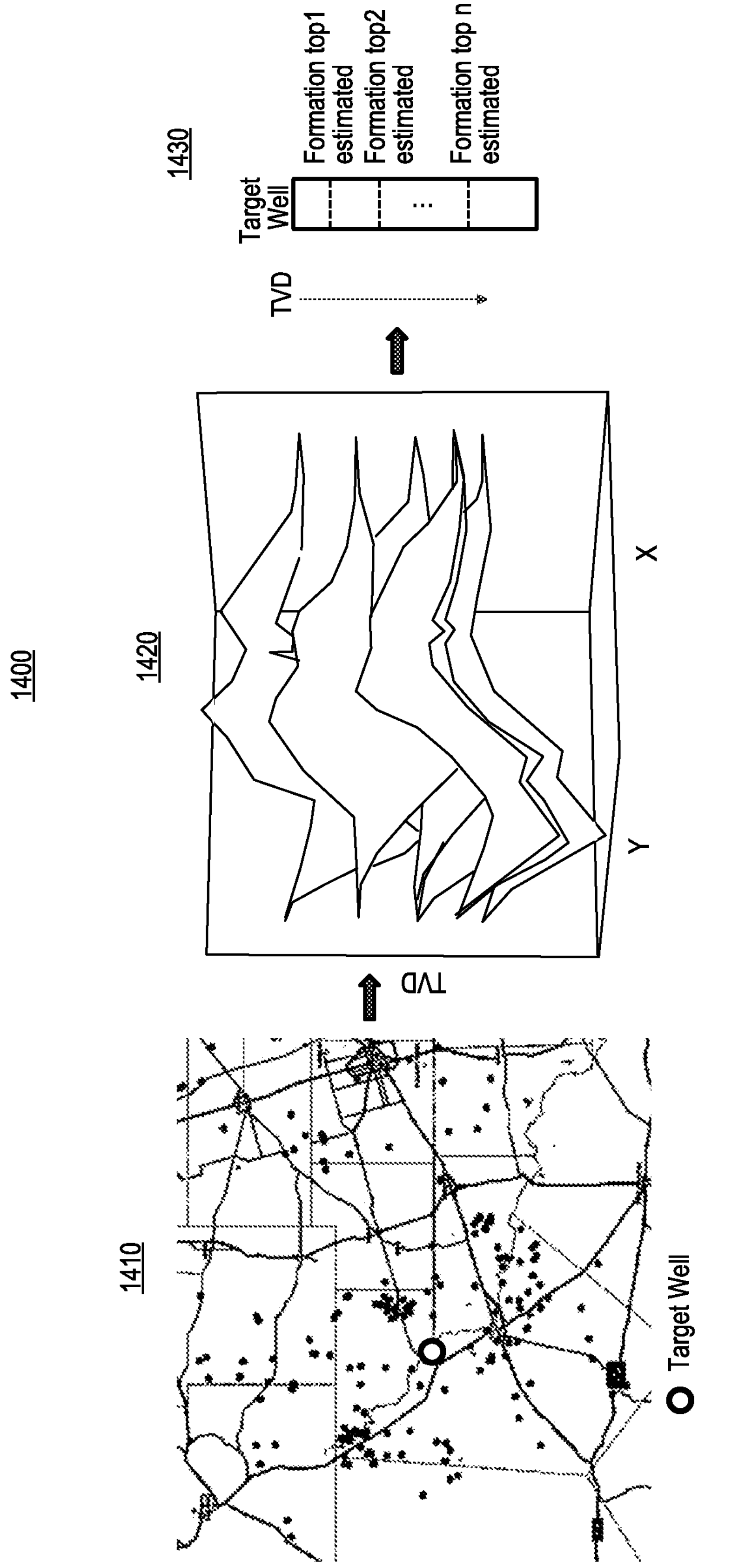
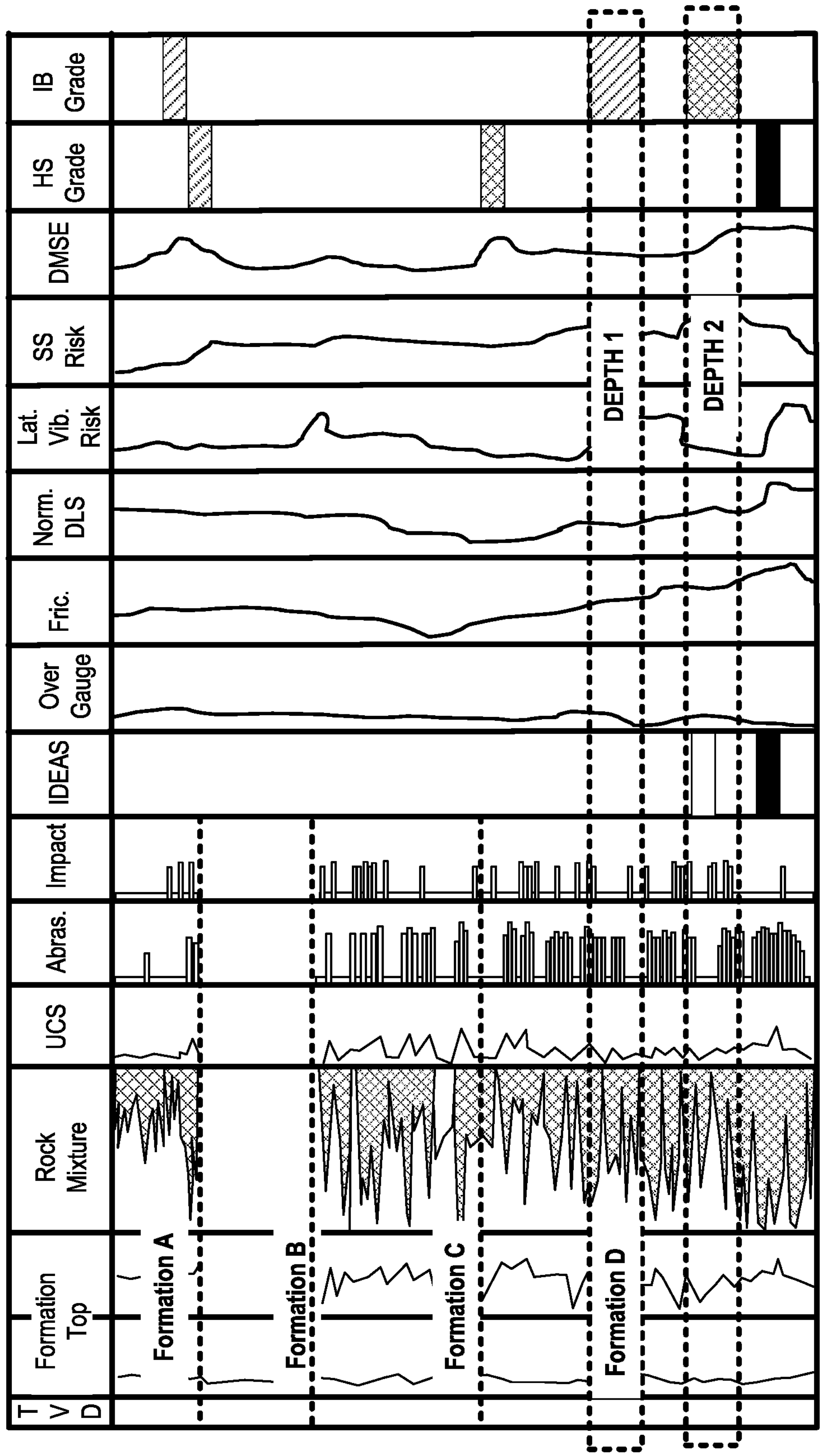

1400
1410
Target Well
TVD
1420
X
Y
TVD
1430
Target Well
...
Formation top1 estimated
Formation top2 estimated
Formation top n estimated

Fig. 14

1500



T V D
Formation Top
Rock Mixture
UCS
Abras.
Impact
IDEAS
Over Gauge
Fric.
Norm. DLS
Lat. Vib. Risk
SS Risk
DMSE
HS Grade
IB Grade
Formation A
Formation B
Formation C
Formation D
DEPTH 1
DEPTH 2

Fig. 15

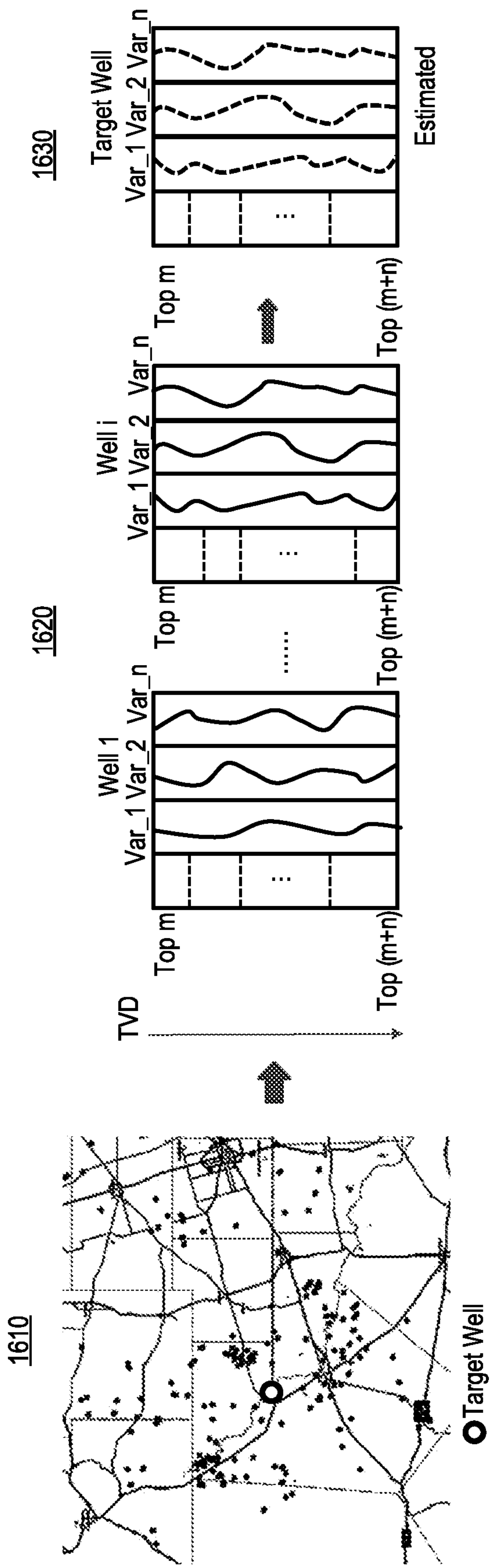

1610
Target Well
TVD
Top m
Top (m+n)
Well 1
Var_1 Var_2 Var_n
1620
Well i
Var_1 Var_2 Var_n
Top m
Top (m+n)
1630
Target Well
Var_1 Var_2 Var_n
Estimated
Top m
Top (m+n)

Fig. 16

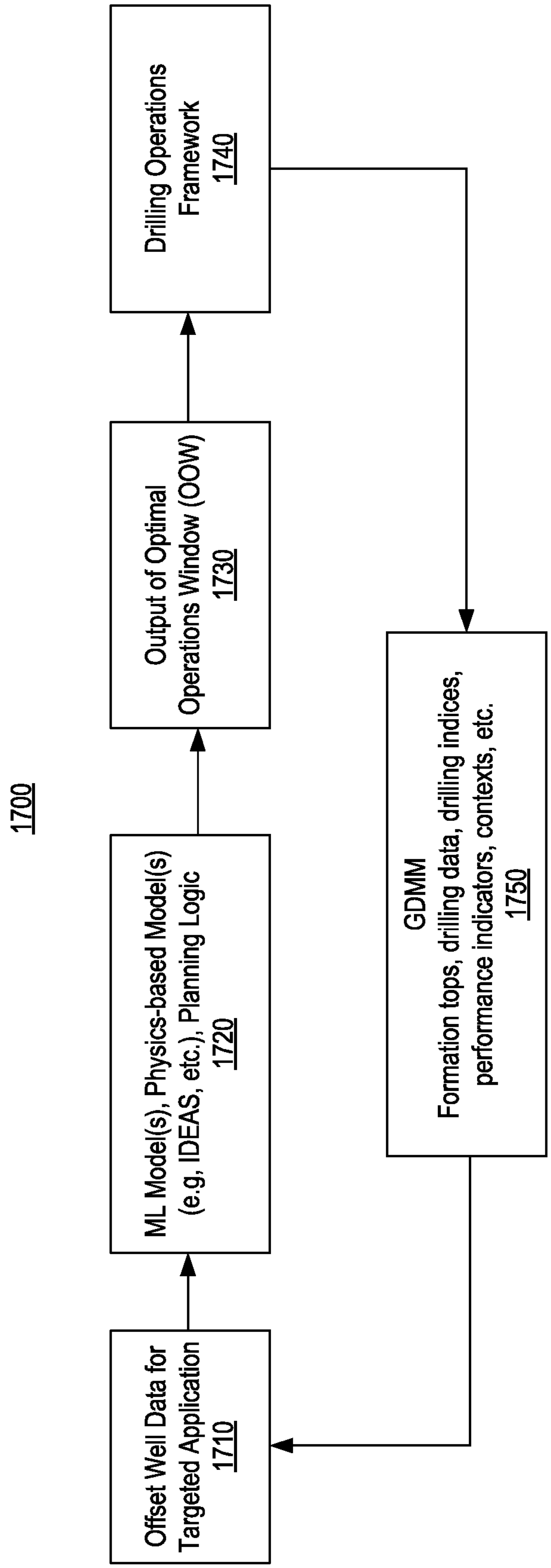
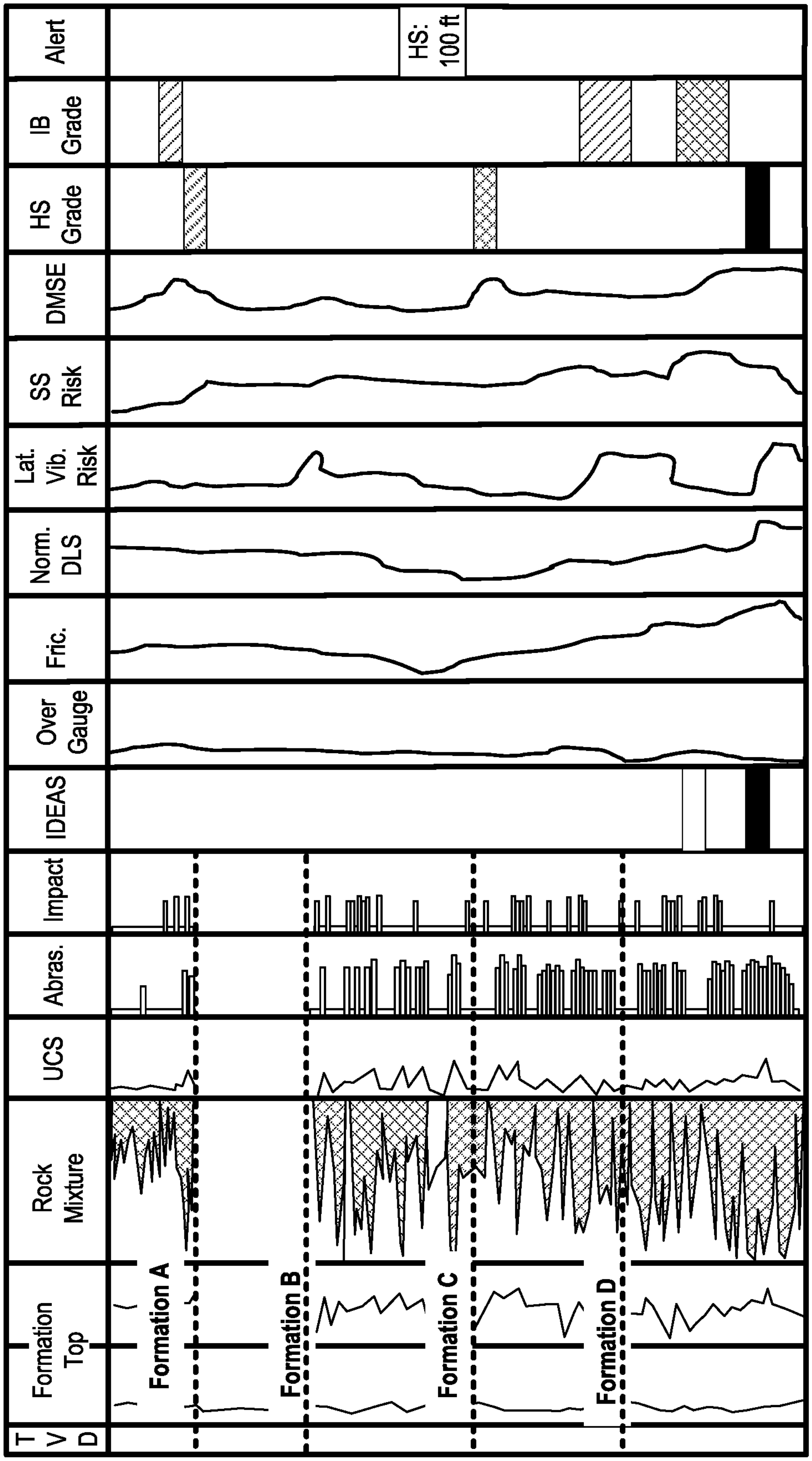

1700
Offset Well Data for Targeted Application
1710
ML Model(s), Physics-based Model(s) (e.g, IDEAS, etc.), Planning Logic
1720
Output of Optimal Operations Window (OOW)
1730
Drilling Operations Framework
1740
GDMM
Formation tops, drilling data, drilling indices, performance indicators, contexts, etc.
1750

Fig. 17

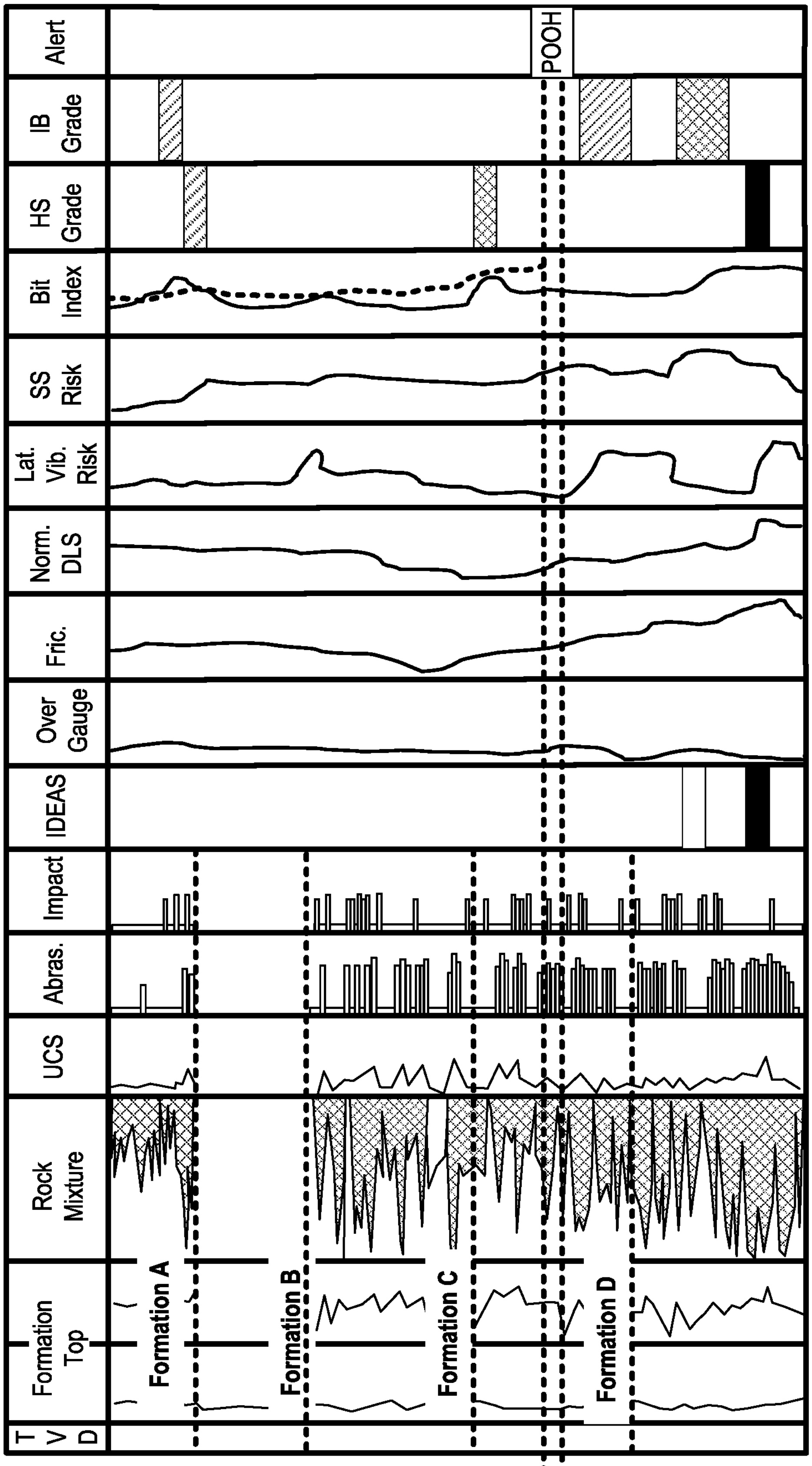
1800



T V D
Formation Top
Rock Mixture
UCS
Abras.
Impact
IDEAS
Over Gauge
Fric.
Norm. DLS
Lat. Vib. Risk
SS Risk
DMSE
HS Grade
IB Grade
Alert
HS: 100 ft
Formation A
Formation B
Formation C
Formation D

Fig. 18

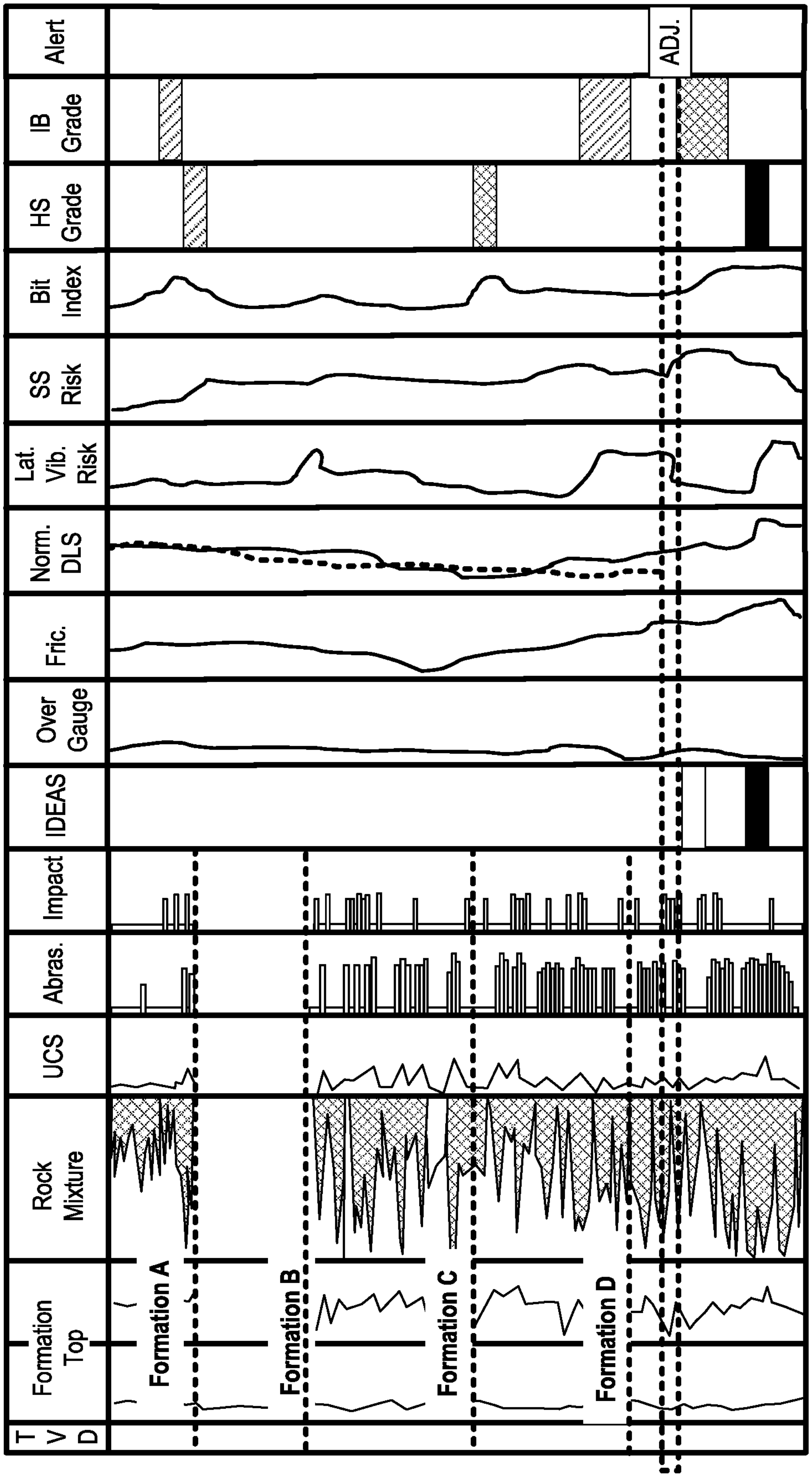
1900



Alert
POOH
IB Grade
HS Grade
Bit Index
SS Risk
Lat. Vib. Risk
Norm. DLS
Fric.
Over Gauge
IDEAS
Impact
Abras.
UCS
Rock Mixture
Formation Top
Formation A
Formation B
Formation C
Formation D
TVD

Fig. 19

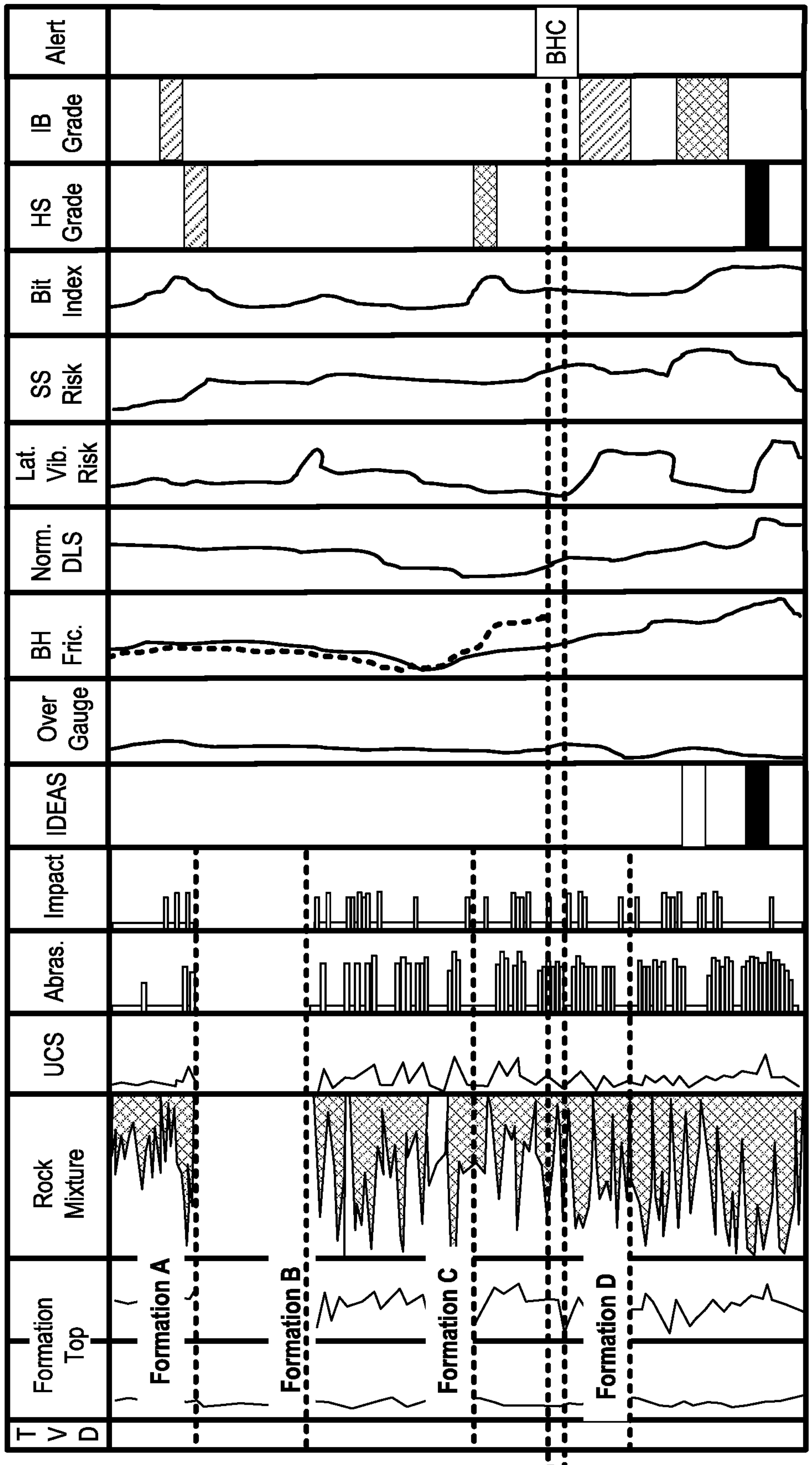
2000



TVD
Formation Top
Formation A
Formation B
Formation C
Formation D
Rock Mixture
UCS
Abras.
Impact
IDEAS
Over Gauge
Fric.
Norm. DLS
Lat. Vib. Risk
SS Risk
Bit Index
HS Grade
IB Grade
Alert
ADJ.

Fig. 20

2100



TVD
Formation Top
Formation A
Formation B
Formation C
Formation D
Rock Mixture
UCS
Abras.
Impact
IDEAS
Over Gauge
BH Fric.
Norm. DLS
Lat. Vib. Risk
SS Risk
Bit Index
HS Grade
IB Grade
Alert
BHC

Fig. 21

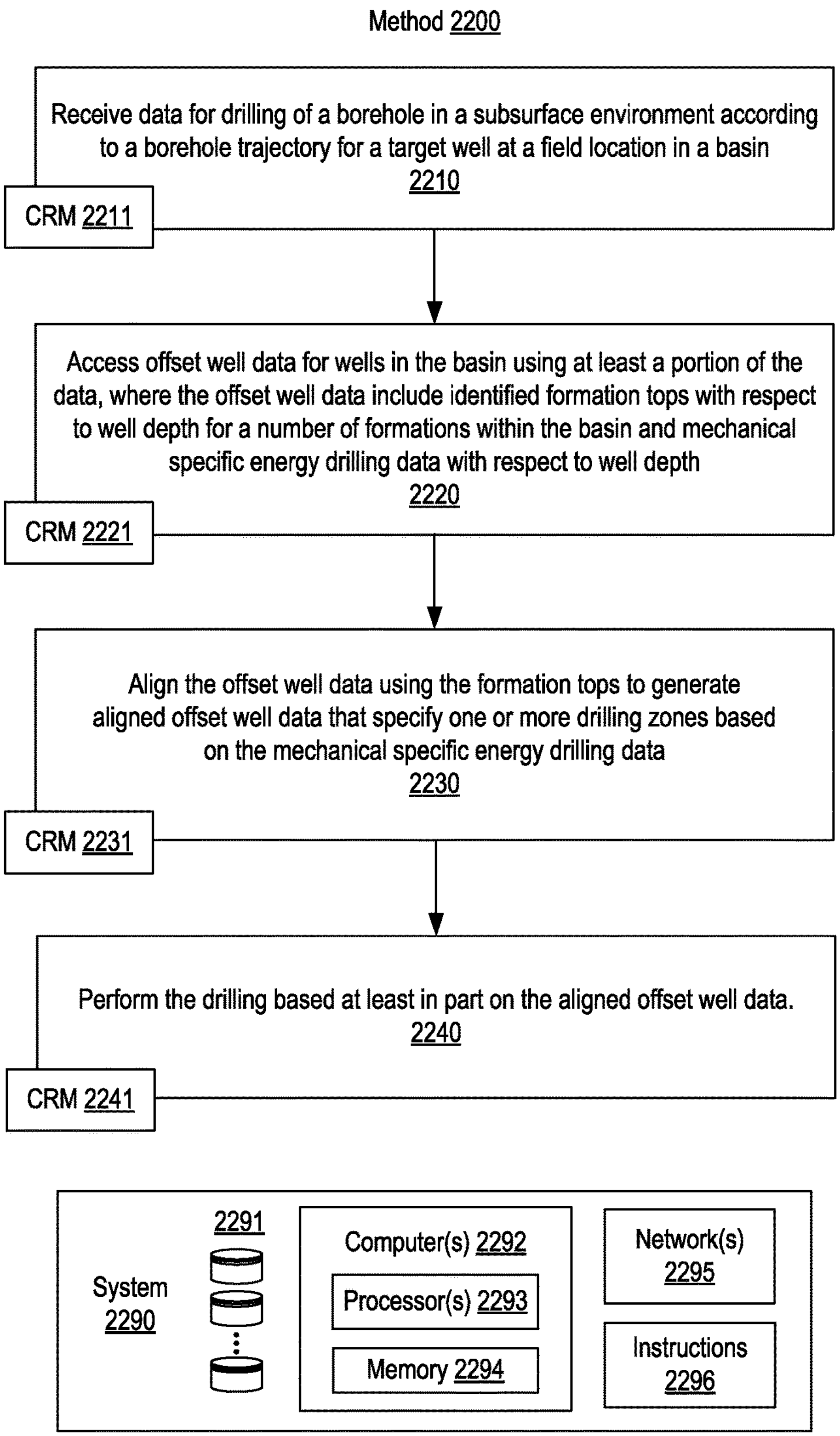

Method 2200
Receive data for drilling of a borehole in a subsurface environment according to a borehole trajectory for a target well at a field location in a basin
2210
CRM 2211
Access offset well data for wells in the basin using at least a portion of the data, where the offset well data include identified formation tops with respect to well depth for a number of formations within the basin and mechanical specific energy drilling data with respect to well depth
2220
CRM 2221
Align the offset well data using the formation tops to generate aligned offset well data that specify one or more drilling zones based on the mechanical specific energy drilling data
2230
CRM 2231
Perform the drilling based at least in part on the aligned offset well data.
2240
CRM 2241
System
2290
2291
Computer(s) 2292
Processor(s) 2293
Memory 2294
Network(s)
2295
Instructions
2296

Fig. 22

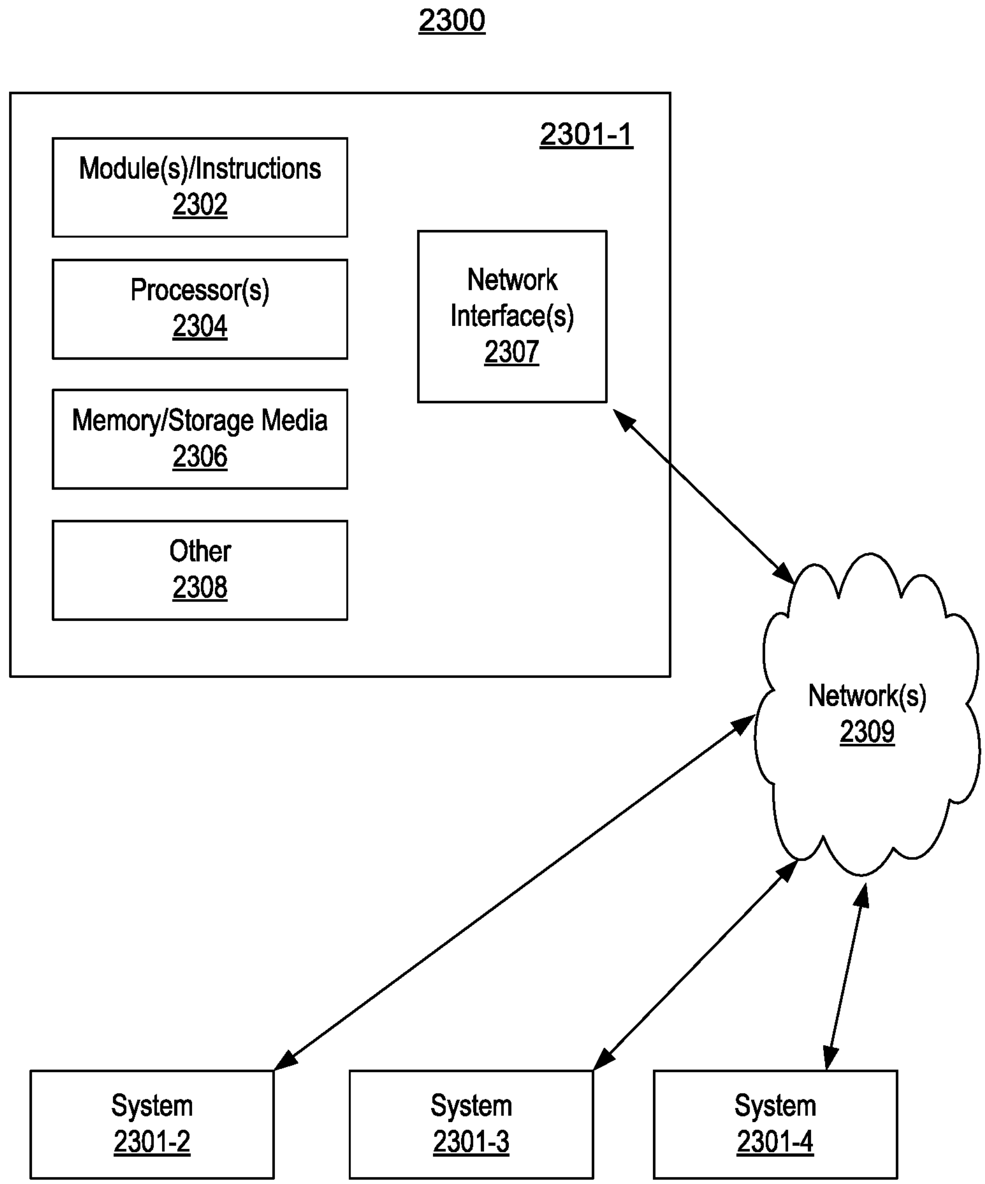

2300
2301-1
Module(s)/Instructions
2302
Processor(s)
2304
Memory/Storage Media
2306
Other
2308
Network
Interface(s)
2307
Network(s)
2309
System
2301-2
System
2301-3
System
2301-4

Fig. 23

DRILLING FRAMEWORK

BACKGROUND

[0001] A reservoir may be a subsurface formation that may be characterized at least in part by its porosity and fluid permeability. As an example, a reservoir may be part of a basin such as a sedimentary basin. A basin may be a depression (e.g., caused by plate tectonic activity, subsidence, etc.) in which sediments accumulate. As an example, where hydrocarbon source rocks occur in combination with appropriate depth and duration of burial, a petroleum system may develop within a basin, which may form a reservoir that includes hydrocarbon fluids (e.g., oil, gas, etc.).

[0002] In oil and gas exploration, interpretation is a process that involves analysis of data to identify and locate various subsurface structures (e.g., horizons, faults, geobodies, etc.) in a geologic environment. Various types of structures (e.g., stratigraphic formations) may be indicative of hydrocarbon traps or flow channels, as may be associated with one or more reservoirs (e.g., fluid reservoirs). In the field of resource extraction, enhancements to interpretation may allow for construction of a more accurate model of a subsurface region, which, in turn, may improve characterization of the subsurface region for purposes of resource extraction. Characterization of one or more subsurface regions in a geologic environment may guide, for example, performance of one or more operations (e.g., field operations, etc.). As an example, a plan may depend on a model of a subsurface region where the plan may specify how a drilling operation may accurately construct a borehole according to a trajectory that penetrates a reservoir, etc., where fluid may be produced via the borehole (e.g., as a completed well, etc.). As an example, one or more workflows may be performed using one or more computational frameworks, systems, etc., for one or more of analysis, acquisition, model building, control, etc., for exploration, interpretation, drilling, fracturing, production, etc.

SUMMARY

[0003] A method may include receiving data for drilling of a borehole in a subsurface environment according to a borehole trajectory for a target well at a field location in a basin; accessing offset well data for wells in the basin using at least a portion of the data, where the offset well data include identified formation tops with respect to well depth for a number of formations within the basin and mechanical specific energy drilling data with respect to well depth; aligning the offset well data using the formation tops to generate aligned offset well data that specify one or more drilling zones based on the mechanical specific energy drilling data; and performing the drilling based at least in part on the aligned offset well data. A system may include one or more processors; memory accessible to at least one of the one or more processors; processor-executable instructions stored in the memory and executable to instruct the system to: receive data for drilling of a borehole in a subsurface environment according to a borehole trajectory for a target well at a field location in a basin; access offset well data for wells in the basin using at least a portion of the data, where the offset well data include identified formation tops with respect to well depth for a number of formations within the basin and mechanical specific energy drilling data with respect to well depth; align the offset well data using the formation tops to generate aligned offset well data that specify one or more drilling zones based on the mechanical specific energy drilling data; and perform the drilling based at least in part on the aligned offset well data. One or more computer-readable storage media may include processor-executable instructions to instruct a computing system to: receive data for drilling of a borehole in a subsurface environment according to a borehole trajectory for a target well at a field location in a basin; access offset well data for wells in the basin using at least a portion of the data, where the offset well data include identified formation tops with respect to well depth for a number of formations within the basin and mechanical specific energy drilling data with respect to well depth; align the offset well data using the formation tops to generate aligned offset well data that specify one or more drilling zones based on the mechanical specific energy drilling data; and perform the drilling based at least in part on the aligned offset well data. Various other apparatuses, systems, methods, etc., are also disclosed.

[0004] This summary is provided to introduce a selection of concepts that are further described below in the detailed description. This summary is not intended to identify key or essential features of the claimed subject matter, nor is it intended to be used as an aid in limiting the scope of the claimed subject matter.

BRIEF DESCRIPTION OF THE DRAWINGS

[0005] The following detailed description refers to the accompanying drawings. Wherever convenient Features and advantages of the described implementations may be more readily understood by reference to the following description taken in conjunction with the accompanying drawings.

[0006] FIG. **1** shows an example of a system;

[0007] FIG. **2** shows an example of a system;

[0008] FIG. **3** shows an example of a system;

[0009] FIG. **4** shows an example of a system;

[0010] FIG. **5** shows an example of a system;

[0011] FIG. **6** shows an example of a system;

[0012] FIG. **7** shows an example of a graphical user interface;

[0013] FIG. **8** shows an example of a workflow;

[0014] FIG. **9** shows an example of a graphical user interface;

[0015] FIG. **10** shows an example of a method;

[0016] FIG. **11** shows an example of a graphical user interface;

[0017] FIG. **12** shows an example of a method;

[0018] FIG. **13** shows an example of a graphical user interface;

[0019] FIG. **14** shows an example of a method;

[0020] FIG. **15** shows an example of a graphical user interface;

[0021] FIG. **16** shows an example of a method;

[0022] FIG. **17** shows an example of a method;

[0023] FIG. **18** shows an example of a graphical user interface;

[0024] FIG. **19** shows an example of a graphical user interface;

[0025] FIG. **20** shows an example of a graphical user interface;

[0026] FIG. **21** shows an example of a graphical user interface;

[0027] FIG. **22** shows an example of a method and an example of a system; and

[0028] FIG. **23** shows an example of a system.

DETAILED DESCRIPTION

[0029] This description is not to be taken in a limiting sense, but rather is made merely for the purpose of describing the general principles of the implementations. The scope of the described implementations should be ascertained with reference to the issued claims.

[0030] FIG. **1** shows an example of a system **100** that includes a workspace framework **110** that may provide for instantiation of, rendering of, interactions with, etc., a graphical user interface (GUI) **120**. In the example of FIG. **1**, the GUI **120** may include graphical controls for computational frameworks (e.g., applications, etc.) **121**, projects **122**, visualization features **123**, one or more other features **124**, data access **125**, and data storage **126**.

[0031] In the example of FIG. **1**, the workspace framework **110** may be tailored to a particular geologic environment such as an example geologic environment **150**. For example, the geologic environment **150** may include layers (e.g., stratification) that include a reservoir **151** and that may be intersected by a fault **153**. As an example, the geologic environment **150** may be outfitted with a variety of sensors, detectors, actuators, etc. For example, equipment **152** may include communication circuitry to receive and to transmit information with respect to one or more networks **155**. Such information may include information associated with downhole equipment **154**, which may be equipment to acquire information, to assist with resource recovery, etc. Other equipment **156** may be located remote from a wellsite and include sensing, detecting, emitting or other circuitry. Such equipment may include storage and communication circuitry to store and to communicate data, instructions, etc. As an example, one or more satellites may be provided for purposes of communications, data acquisition, etc. For example, FIG. **1** shows a satellite in communication with the network **155** that may be configured for communications, noting that the satellite may additionally or alternatively include circuitry for imagery (e.g., spatial, spectral, temporal, radiometric, etc.).

[0032] FIG. **1** also shows the geologic environment **150** as optionally including equipment **157** and **158** associated with a well that includes a substantially horizontal portion that may intersect with one or more fractures **159**. For example, consider a well in a shale formation that may include natural fractures, artificial fractures (e.g., hydraulic fractures) or a combination of natural and artificial fractures. As an example, a well may be drilled for a reservoir that is laterally extensive. In such an example, lateral variations in properties, stresses, etc. may exist where an assessment of such variations may assist with planning, operations, etc. to develop a laterally extensive reservoir (e.g., via fracturing, injecting, extracting, etc.). As an example, the equipment **157** and/or **158** may include components, a system, systems, etc. for fracturing, seismic sensing, analysis of seismic data, assessment of one or more fractures, etc.

[0033] In the example of FIG. **1**, the GUI **120** shows some examples of computational frameworks, including the DRILLPLAN, DRILLOPS, PETREL, TECHLOG, PETROMOD, ECLIPSE, PIPESIM, and INTERSECT frameworks (SLB, Houston, Texas).

[0034] The DRILLPLAN framework provides for digital well construction planning and includes features for automation of repetitive tasks and validation workflows, enabling improved quality drilling programs (e.g., digital drilling plans, etc.) to be produced quickly with assured coherency.

[0035] The DRILLOPS framework may execute a digital drilling plan and ensures plan adherence, while delivering goal-based automation. The DRILLOPS framework may generate activity plans automatically individual operations, whether they are monitored and/or controlled on the rig or in town. Automation may utilize data analysis and learning systems to assist and optimize tasks, such as, for example, setting ROP to drilling a stand. A preset menu of automatable drilling tasks may be rendered, and, using data analysis and models, a plan may be executed in a manner to achieve a specified goal, where, for example, measurements may be utilized for calibration. The DRILLOPS framework provides flexibility to modify and replan activities dynamically, for example, based on a live appraisal of various factors (e.g., equipment, personnel, and supplies). Well construction activities (e.g., tripping, drilling, cementing, etc.) may be continually monitored and dynamically updated using feedback from operational activities. The DRILLOPS framework may provide for various levels of automation based on planning and/or re-planning (e.g., via the DRILLPLAN framework), feedback, etc.

[0036] The PETREL framework may be part of the DELFI environment for utilization in geosciences and geoengineering, for example, to analyze subsurface data from exploration to production of fluid from a reservoir. The DELFI cognitive exploration and production (E&P) environment (SLB, Houston, Texas), referred to herein as the DELFI environment or DELFI framework, is a secure, cognitive, cloud-based collaborative environment that integrates data and workflows with digital technologies, such as artificial intelligence and machine learning.

[0037] The PETREL framework provides components that allow for optimization of various exploration, development and production operations. The PETREL framework includes seismic to simulation software components that may output information for use in increasing reservoir performance, for example, by improving asset team productivity. Through use of such a framework, various professionals (e.g., geophysicists, geologists, and reservoir engineers) may develop collaborative workflows and integrate operations to streamline processes (e.g., with respect to one or more geologic environments, etc.). Such a framework may be considered an application (e.g., executable using one or more devices) and may be considered a data-driven application (e.g., where data is input for purposes of modeling, simulating, etc.).

[0038] The TECHLOG framework may handle and process field and laboratory data for a variety of geologic environments (e.g., deepwater exploration, shale, etc.). The TECHLOG framework may structure wellbore data for analyses, planning, etc.

[0039] The PETROMOD framework provides petroleum systems modeling capabilities that may combine one or more of seismic, well, and geological information to model the evolution of a sedimentary basin. The PETROMOD framework may predict if, and how, a reservoir has been charged with hydrocarbons, including the source and timing of hydrocarbon generation, migration routes, quantities, and hydrocarbon type in the subsurface or at surface conditions.

[0040] The ECLIPSE framework provides a reservoir simulator (e.g., as a computational framework) with numerical solutions for fast and accurate prediction of dynamic behavior for various types of reservoirs and development schemes.

[0041] The INTERSECT framework provides a high-resolution reservoir simulator for simulation of detailed geological features and quantification of uncertainties, for example, by creating accurate production scenarios and, with the integration of precise models of the surface facilities and field operations, the INTERSECT framework may produce reliable results, which may be continuously updated by real-time data exchanges (e.g., from one or more types of data acquisition equipment in the field that may acquire data during one or more types of field operations, etc.). The INTERSECT framework may provide completion configurations for complex wells where such configurations may be built in the field, may provide detailed enhanced-oil-recovery (EOR) formulations where such formulations may be implemented in the field, may analyze application of steam injection and other thermal EOR techniques for implementation in the field, advanced production controls in terms of reservoir coupling and flexible field management, and flexibility to script customized solutions for improved modeling and field management control. The INTERSECT framework, as with the other example frameworks, may be utilized as part of the DELFI environment, for example, for rapid simulation of multiple concurrent cases. For example, a workflow may utilize one or more of the DELFI environment on demand reservoir simulation features.

[0042] The aforementioned DELFI environment provides various features for workflows as to subsurface analysis, planning, construction and production, for example, as illustrated in the workspace framework **110**. As shown in FIG. **1**, outputs from the workspace framework **110** may be utilized for directing, controlling, etc., one or more processes in the geologic environment **150** and, feedback **160**, may be received via one or more interfaces in one or more forms (e.g., acquired data as to operational conditions, equipment conditions, environment conditions, etc.).

[0043] As an example, a workflow may progress to a geology and geophysics ("G&G") service provider, which may generate a well trajectory, which may involve execution of one or more G&G frameworks (e.g., consider the PETREL framework, etc.).

[0044] In the example of FIG. **1**, the visualization features **123** may be implemented via the workspace framework **110**, for example, to perform tasks as associated with one or more of subsurface regions, planning operations, constructing wells and/or surface fluid networks, and producing from a reservoir.

[0045] As an example, a visualization process may implement one or more of various features that may be suitable for one or more web applications. For example, a template may involve use of the JAVASCRIPT object notation format (JSON) and/or one or more other languages/formats. As an example, a framework may include one or more converters. For example, consider a JSON to PYTHON converter and/or a PYTHON to JSON converter. Such an approach may provide for compatibility of devices, frameworks, etc., with respect to one or more sets of instructions.

[0046] As an example, visualization features may provide for visualization of various earth models, properties, etc., in one or more dimensions. As an example, visualization features may provide for rendering of information in multiple dimensions, which may optionally include multiple resolution rendering. In such an example, information being rendered may be associated with one or more frameworks and/or one or more data stores. As an example, visualization features may include one or more control features for control of equipment, which may include, for example, field equipment that may perform one or more field operations. As an example, a workflow may utilize one or more frameworks to generate information that may be utilized to control one or more types of field equipment (e.g., drilling equipment, wireline equipment, fracturing equipment, etc.).

[0047] As to a reservoir model that may be suitable for utilization by a simulator, consider acquisition of seismic data as acquired via reflection seismology, which finds use in geophysics, for example, to estimate properties of subsurface formations. As an example, reflection seismology may provide seismic data representing waves of elastic energy (e.g., as transmitted by P-waves and S-waves, in a frequency range of approximately 1 Hz to approximately 100 Hz). Seismic data may be processed and interpreted, for example, to understand better composition, fluid content, extent and geometry of subsurface rocks. Such interpretation results may be utilized to plan, simulate, perform, etc., one or more operations for production of fluid from a reservoir (e.g., reservoir rock, etc.).

[0048] As an example, a model may be a simulated version of a geologic environment. As an example, a simulator may include features for simulating physical phenomena in a geologic environment based at least in part on a model or models. A simulator, such as a reservoir simulator, may simulate fluid flow in a geologic environment based at least in part on a model that may be generated via a framework that receives seismic data. A simulator may be a computerized system (e.g., a computing system) that may execute instructions using one or more processors to solve a system of equations that describe physical phenomena subject to various constraints. In such an example, the system of equations may be spatially defined (e.g., numerically discretized) according to a spatial model that that includes layers of rock, geobodies, etc., that have corresponding positions that may be based on interpretation of seismic and/or other data. A spatial model may be a cell-based model where cells are defined by a grid (e.g., a mesh). A cell in a cell-based model may represent a physical area or volume in a geologic environment where the cell may be assigned physical properties (e.g., permeability, fluid properties, etc.) that may be germane to one or more physical phenomena (e.g., fluid volume, fluid flow, pressure, etc.). A reservoir simulation model may be a spatial model that may be cell-based.

[0049] While several simulators are illustrated in the example of FIG. **1**, one or more other simulators may be utilized, additionally or alternatively. For example, consider the VISAGE geomechanics simulator (SLB, Houston Texas) or the PIPESIM network simulator (SLB, Houston Texas), etc.

[0050] As an example, a workflow may utilize one or more types of data for one or more processes (e.g., stratigraphic modeling, basin modeling, completion designs, drilling, production, injection, etc.). As an example, one or more tools may provide data that may be used in a workflow or workflows that may implement one or more frameworks (e.g., PETREL, TECHLOG, PETROMOD, ECLIPSE, etc.).

[0051] In the example of FIG. **1**, drilling may be performed in the geologic environment **150**, for example, to access the reservoir **151**, which may be accessed from land or offshore. In FIG. **1**, the downhole equipment **154** may be, for example, part of a bottom hole assembly (BHA). The BHA may be used to drill a well. The downhole equipment **154** may communicate information to equipment at the surface and may receive instructions and information from the equipment at the surface. During a well construction process, a variety of operations (such as cementing, wireline evaluation, testing, etc.) may be conducted. In such embodiments, data collected by tools and sensors and used for reasons such as reservoir characterization may be collected and transmitted.

[0052] A well may include a substantially horizontal portion (e.g., lateral portion) that may intersect with one or more fractures. For example, a well in a shale formation may pass through natural fractures, artificial fractures (e.g., hydraulic fractures), or a combination thereof. Such a well may be constructed using directional drilling techniques as described herein. However, these same techniques may be used in connection with other types of directional wells (such as slant wells, S-shaped wells, deep inclined wells, and others) and are not limited to horizontal wells.

[0053] FIG. **2** shows an example of a wellsite system **200** (e.g., at a wellsite that may be onshore or offshore). As shown, the wellsite system **200** may include a mud tank **201** for holding mud and other material (e.g., where mud may be a drilling fluid), a suction line **203** that serves as an inlet to a mud pump **204** for pumping mud from the mud tank **201** such that mud flows to a vibrating hose **206**, a drawworks **207** for winching drill line or drill lines **212**, a standpipe **208** that receives mud from the vibrating hose **206**, a kelly hose **209** that receives mud from the standpipe **208**, a gooseneck or goosenecks **210**, a traveling block **211**, a crown block **213** for carrying the traveling block **211** via the drill line or drill lines **212**, a derrick **214**, a kelly **218** or a top drive **240**, a kelly drive bushing **219**, a rotary table **220**, a drill floor **221**, a bell nipple **222**, one or more blowout preventors (BOPs) **223**, a drillstring **225**, a drill bit **226**, a casing head **227** and a flow pipe **228** that carries mud and other material to, for example, the mud tank **201**.

[0054] In the example system of FIG. **2**, a borehole **232** is formed in subsurface formations **230** by rotary drilling; noting that various example embodiments may also use one or more directional drilling techniques, equipment, etc.

[0055] As shown in the example of FIG. **2**, the drillstring **225** is suspended within the borehole **232** and has a drillstring assembly **250** that includes the drill bit **226** at its lower end. As an example, the drillstring assembly **250** may be a bottom hole assembly (BHA).

[0056] The wellsite system **200** may provide for operation of the drillstring **225** and other operations. As shown, the wellsite system **200** includes the traveling block **211** and the derrick **214** positioned over the borehole **232**. As mentioned, the wellsite system **200** may include the rotary table **220** where the drillstring **225** pass through an opening in the rotary table **220**.

[0057] As shown in the example of FIG. **2**, the wellsite system **200** may include the kelly **218** and associated components, etc., or a top drive **240** and associated components. As to a kelly example, the kelly **218** may be a square or hexagonal metal/alloy bar with a hole drilled therein that serves as a mud flow path. The kelly **218** may be used to transmit rotary motion from the rotary table **220** via the kelly drive bushing **219** to the drillstring **225**, while allowing the drillstring **225** to be lowered or raised during rotation. The kelly **218** may pass through the kelly drive bushing **219**, which may be driven by the rotary table **220**. As an example, the rotary table **220** may include a master bushing that operatively couples to the kelly drive bushing **219** such that rotation of the rotary table **220** may turn the kelly drive bushing **219** and hence the kelly **218**. The kelly drive bushing **219** may include an inside profile matching an outside profile (e.g., square, hexagonal, etc.) of the kelly **218**; however, with slightly larger dimensions so that the kelly **218** may freely move up and down inside the kelly drive bushing **219**.

[0058] As to a top drive example, the top drive **240** may provide functions performed by a kelly and a rotary table. The top drive **240** may turn the drillstring **225**. As an example, the top drive **240** may include one or more motors (e.g., electric and/or hydraulic) connected with appropriate gearing to a short section of pipe called a quill, that in turn may be screwed into a saver sub or the drillstring **225** itself. The top drive **240** may be suspended from the traveling block **211**, so the rotary mechanism is free to travel up and down the derrick **214**. As an example, a top drive **240** may allow for drilling to be performed with more joint stands than a kelly/rotary table approach.

[0059] In the example of FIG. **2**, the mud tank **201** may hold mud, which may be one or more types of drilling fluids. As an example, a wellbore may be drilled to produce fluid, inject fluid or both (e.g., hydrocarbons, minerals, water, etc.).

[0060] In the example of FIG. **2**, the drillstring **225** (e.g., including one or more downhole tools) may be composed of a series of pipes threadably connected together to form a long tube with the drill bit **226** at the lower end thereof. As the drillstring **225** is advanced into a wellbore for drilling, at some point in time prior to or coincident with drilling, the mud may be pumped by the pump **204** from the mud tank **201** (e.g., or other source) via the lines **206**, **208** and **209** to a port of the kelly **218** or, for example, to a port of the top drive **240**. The mud may then flow via a passage (e.g., or passages) in the drillstring **225** and out of ports located on the drill bit **226** (see, e.g., a directional arrow). As the mud exits the drillstring **225** via ports in the drill bit **226**, it may then circulate upwardly through an annular region between an outer surface(s) of the drillstring **225** and surrounding wall(s) (e.g., open borehole, casing, etc.), as indicated by directional arrows. In such a manner, the mud lubricates the drill bit **226** and carries heat energy (e.g., frictional or other energy) and formation cuttings to the surface where the mud (e.g., and cuttings) may be returned to the mud tank **201**, for example, for recirculation (e.g., with processing to remove cuttings, etc.).

[0061] The mud pumped by the pump **204** into the drillstring **225** may, after exiting the drillstring **225**, form a mudcake that lines the wellbore which, among other functions, may reduce friction between the drillstring **225** and surrounding wall(s) (e.g., borehole, casing, etc.). A reduction in friction may facilitate advancing or retracting the drillstring **225**. During a drilling operation, the entire drillstring **225** may be pulled from a wellbore and optionally replaced, for example, with a new or sharpened drill bit, a smaller diameter drillstring, etc. As mentioned, the act of pulling a drillstring out of a hole or replacing it in a hole is referred to as tripping. A trip may be referred to as an upward trip or an outward trip or as a downward trip or an inward trip depending on trip direction.

[0062] As an example, consider a downward trip where upon arrival of the drill bit **226** of the drillstring **225** at a bottom of a wellbore, pumping of the mud commences to lubricate the drill bit **226** for purposes of drilling to enlarge the wellbore. As mentioned, the mud may be pumped by the pump **204** into a passage of the drillstring **225** and, upon filling of the passage, the mud may be used as a transmission medium to transmit energy, for example, energy that may encode information as in mud-pulse telemetry.

[0063] As an example, mud-pulse telemetry equipment may include a downhole device configured to effect changes in pressure in the mud to create an acoustic wave or waves upon which information may modulated. In such an example, information from downhole equipment (e.g., one or more modules of the drillstring **225**) may be transmitted uphole to an uphole device, which may relay such information to other equipment for processing, control, etc.

[0064] As an example, telemetry equipment may operate via transmission of energy via the drillstring **225** itself. For example, consider a signal generator that imparts coded energy signals to the drillstring **225** and repeaters that may receive such energy and repeat it to further transmit the coded energy signals (e.g., information, etc.).

[0065] As an example, the drillstring **225** may be fitted with telemetry equipment **252** that includes a rotatable drive shaft, a turbine impeller mechanically coupled to the drive shaft such that the mud may cause the turbine impeller to rotate, a modulator rotor mechanically coupled to the drive shaft such that rotation of the turbine impeller causes said modulator rotor to rotate, a modulator stator mounted adjacent to or proximate to the modulator rotor such that rotation of the modulator rotor relative to the modulator stator creates pressure pulses in the mud, and a controllable brake for selectively braking rotation of the modulator rotor to modulate pressure pulses. In such an example, an alternator may be coupled to the aforementioned drive shaft where the alternator includes at least one stator winding electrically coupled to a control circuit to selectively short the at least one stator winding to electromagnetically brake the alternator and thereby selectively brake rotation of the modulator rotor to modulate the pressure pulses in the mud.

[0066] In the example of FIG. **2**, an uphole control and/or data acquisition system **262** may include circuitry to sense pressure pulses generated by telemetry equipment **252** and, for example, communicate sensed pressure pulses or information derived therefrom for process, control, etc.

[0067] The assembly **250** of the illustrated example includes a logging-while-drilling (LWD) module **254**, a measurement-while-drilling (MWD) module **256**, an optional module **258**, a rotary-steerable system (RSS) and/or motor **260**, and the drill bit **226**. Such components or modules may be referred to as tools where a drillstring may include a plurality of tools.

[0068] As to an RSS, it involves technology utilized for directional drilling. Directional drilling involves drilling into the Earth to form a deviated bore such that the trajectory of the bore is not vertical; rather, the trajectory deviates from vertical along one or more portions of the bore. As an example, consider a target that is located at a lateral distance from a surface location where a rig may be stationed. In such an example, drilling may commence with a vertical portion and then deviate from vertical such that the bore is aimed at the target and, eventually, reaches the target. Directional drilling may be implemented where a target may be inaccessible from a vertical location at the surface of the Earth, where material exists in the Earth that may impede drilling or otherwise be detrimental (e.g., consider a salt dome, etc.), where a formation is laterally extensive (e.g., consider a relatively thin yet laterally extensive reservoir), where multiple bores are to be drilled from a single surface bore, where a relief well is desired, etc.

[0069] One approach to directional drilling involves a mud motor; however, a mud motor may present some challenges depending on factors such as rate of penetration (ROP), transferring weight to a bit (e.g., weight on bit, WOB) due to friction, etc. A mud motor may be a positive displacement motor (PDM) that operates to drive a bit (e.g., during directional drilling, etc.). A PDM operates as drilling fluid is pumped through it where the PDM converts hydraulic power of the drilling fluid into mechanical power to cause the bit to rotate.

[0070] As an example, a mud motor (e.g., PDM) may be operated in different modes, which may include a rotating mode and a sliding mode. A sliding mode involves drilling with a mud motor rotating the bit downhole without rotating the drillstring from the surface. Such an operation may be conducted when a BHA has been fitted with a bent sub or a bent housing mud motor, or both, for directional drilling. Sliding may be used in building and controlling or adjusting hole angle. In directional drilling, pointing of a bit may be accomplished through a bent sub, which may have a relatively small angle offset from the axis of a drillstring, and a measurement device to determine the direction of offset. Without turning the drillstring, the bit may be rotated with mud flow through the mud motor to drill in the direction it is pointed. With steerable motors, when a desired wellbore direction is attained, the entire drillstring may be rotated to drill straight rather than at an angle. By controlling the amount of hole drilled in the sliding mode versus the rotating mode, a wellbore trajectory may be controlled rather precisely.

[0071] As an example, a PDM may operate in a combined rotating mode where surface equipment is utilized to rotate a bit of a drillstring (e.g., a rotary table, a top drive, etc.) by rotating the entire drillstring and where drilling fluid is utilized to rotate the bit of the drillstring. In such an example, a surface RPM (SRPM) may be determined by use of the surface equipment and a downhole RPM of the mud motor may be determined using various factors related to flow of drilling fluid, mud motor type, etc. As an example, in the combined rotating mode, bit RPM may be determined or estimated as a sum of the SRPM and the mud motor RPM, assuming the SRPM and the mud motor RPM are in the same direction.

[0072] As an example, a PDM mud motor may operate in a so-called sliding mode, when the drillstring is not rotated from the surface. In such an example, a bit RPM may be determined or estimated based on the RPM of the mud motor.

[0073] An RSS may drill directionally where there is continuous rotation from surface equipment, which may alleviate the sliding of a steerable motor (e.g., a PDM). An RSS may be deployed when drilling directionally (e.g., deviated, horizontal, or extended-reach wells). An RSS may aim to minimize interaction with a borehole wall, which may help to preserve borehole quality. An RSS may aim to exert a relatively consistent side force akin to stabilizers that rotate with the drillstring or orient the bit in the desired direction while continuously rotating at the same number of rotations per minute as the drillstring.

[0074] The LWD module **254** may be housed in a suitable type of drill collar and may contain one or a plurality of selected types of logging tools. It will also be understood that more than one LWD and/or MWD module may be employed. Where the position of an LWD module is mentioned, as an example, it may refer to a module at the position of the LWD module **254**, the MWD module **256**, etc. An LWD module may include capabilities for measuring, processing, and storing information, as well as for communicating with the surface equipment. In the illustrated example, the LWD module **254** may include a seismic measuring device.

[0075] The MWD module **256** may be housed in a suitable type of drill collar and may contain one or more devices for measuring characteristics of the drillstring **225** and the drill bit **226**. As an example, the MWD module **256** may include equipment for generating electrical power, for example, to power various components of the drillstring **225**. As an example, the MWD module **256** may include the telemetry equipment **252**, for example, where the turbine impeller may generate power by flow of the mud; it being understood that other power and/or battery systems may be employed for purposes of powering various components. As an example, the MWD module **256** may include one or more of the following types of measuring devices: a weight-on-bit measuring device, a torque measuring device, a vibration measuring device, a shock measuring device, a stick slip measuring device, a direction measuring device, and an inclination measuring device.

[0076] FIG. **2** also shows some examples of types of holes that may be drilled. For example, consider a slant hole **272**, an S-shaped hole **274**, a deep inclined hole **276** and a horizontal hole **278**.

[0077] As an example, a drilling operation may include directional drilling where, for example, at least a portion of a well includes a curved axis. For example, consider a radius that defines curvature where an inclination with regard to the vertical may vary until reaching an angle between about 30 degrees and about 60 degrees or, for example, an angle to about 90 degrees or possibly greater than about 90 degrees.

[0078] As an example, a directional well may include several shapes where each of the shapes may aim to meet particular operational demands. As an example, a drilling process may be performed on the basis of information as and when it is relayed to a drilling engineer. As an example, inclination and/or direction may be modified based on information received during a drilling process.

[0079] As an example, deviation of a bore may be accomplished in part by use of a downhole motor and/or a turbine. As to a motor, for example, a drillstring may include a positive displacement motor (PDM).

[0080] As an example, a system may be a steerable system and include equipment to perform method such as geosteering. As mentioned, a steerable system may be or include an RSS. As an example, a steerable system may include a PDM or of a turbine on a lower part of a drillstring which, just above a drill bit, a bent sub may be mounted. As an example, above a PDM, MWD equipment that provides real time or near real time data of interest (e.g., inclination, direction, pressure, temperature, real weight on the drill bit, torque stress, etc.) and/or LWD equipment may be installed. As to the latter, LWD equipment may make it possible to send to the surface various types of data of interest, including for example, geological data (e.g., gamma ray log, resistivity, density and sonic logs, etc.).

[0081] The coupling of sensors providing information on the course of a well trajectory, in real time or near real time, with, for example, one or more logs characterizing the formations from a geological viewpoint, may allow for implementing a geosteering method. Such a method may include navigating a subsurface environment, for example, to follow a desired route to reach a desired target or targets.

[0082] As an example, a drillstring may include an azimuthal density neutron (ADN) tool for measuring density and porosity; a MWD tool for measuring inclination, azimuth and shocks; a compensated dual resistivity (CDR) tool for measuring resistivity and gamma ray related phenomena; one or more variable gauge stabilizers; one or more bend joints; and a geosteering tool, which may include a motor and optionally equipment for measuring and/or responding to one or more of inclination, resistivity and gamma ray related phenomena.

[0083] As an example, geosteering may include intentional directional control of a wellbore based on results of downhole geological logging measurements in a manner that aims to keep a directional wellbore within a desired region, zone (e.g., a pay zone), etc. As an example, geosteering may include directing a wellbore to keep the wellbore in a particular section of a reservoir, for example, to minimize gas and/or water breakthrough and, for example, to maximize economic production from a well that includes the wellbore.

[0084] Referring again to FIG. **2**, the wellsite system **200** may include one or more sensors **264** that are operatively coupled to the control and/or data acquisition system **262**. As an example, a sensor or sensors may be at surface locations. As an example, a sensor or sensors may be at downhole locations. As an example, a sensor or sensors may be at one or more remote locations that are not within a distance of the order of about one hundred meters from the wellsite system **200**. As an example, a sensor or sensor may be at an offset wellsite where the wellsite system **200** and the offset wellsite are in a common field (e.g., oil and/or gas field).

[0085] As an example, one or more of the sensors **264** may be provided for tracking pipe, tracking movement of at least a portion of a drillstring, etc.

[0086] As an example, the system **200** may include one or more sensors **266** that may sense and/or transmit signals to a fluid conduit such as a drilling fluid conduit (e.g., a drilling mud conduit). For example, in the system **200**, the one or more sensors **266** may be operatively coupled to portions of the standpipe **208** through which mud flows. As an example, a downhole tool may generate pulses that may travel through the mud and be sensed by one or more of the one or more sensors **266**. In such an example, the downhole tool may include associated circuitry such as, for example, encoding circuitry that may encode signals, for example, to reduce demands as to transmission. As an example, circuitry at the surface may include decoding circuitry to decode encoded information transmitted at least in part via mud-pulse telemetry. As an example, circuitry at the surface may include encoder circuitry and/or decoder circuitry and circuitry downhole may include encoder circuitry and/or decoder circuitry. As an example, the system **200** may include a transmitter that may generate signals that may be transmitted downhole via mud (e.g., drilling fluid) as a transmission medium.

[0087] As an example, one or more portions of a drillstring may become stuck. The term stuck may refer to one or more of varying degrees of inability to move or remove a drill-string from a bore. As an example, in a stuck condition, it might be possible to rotate pipe or lower it back into a bore or, for example, in a stuck condition, there may be an inability to move the drillstring axially in the bore, though some amount of rotation may be possible. As an example, in a stuck condition, there may be an inability to move at least a portion of the drillstring axially and rotationally.

[0088] As to the term "stuck pipe", this may refer to a portion of a drillstring that cannot be rotated or moved axially. As an example, a condition referred to as "differential sticking" may be a condition whereby the drillstring cannot be moved (e.g., rotated or reciprocated) along the axis of the bore. Differential sticking may occur when high-contact forces caused by low reservoir pressures, high wellbore pressures, or both, are exerted over a sufficiently large area of the drillstring. Differential sticking may have time and financial cost.

[0089] As an example, a sticking force may be a product of the differential pressure between the wellbore and the reservoir and the area that the differential pressure is acting upon. This means that a relatively low differential pressure (delta p) applied over a large working area may be just as effective in sticking pipe as may a high differential pressure applied over a small area.

[0090] As an example, a condition referred to as "mechanical sticking" may be a condition where limiting or prevention of motion of the drillstring by a mechanism other than differential pressure sticking occurs. Mechanical sticking may be caused, for example, by one or more of junk in the hole, wellbore geometry anomalies, cement, keyseats or a buildup of cuttings in the annulus.

[0091] As explained, a wellsite system may include various types of equipment for handling fluid such as, for example, drilling fluid (e.g., mud). As explained, drilling fluid may provide one or more functions (e.g., lubrication, transport of cutting, etc.).

[0092] Drilling fluid may be composed of a number of liquid and/or gaseous fluids and mixtures of fluids and solids (e.g., as solid suspensions, mixtures and emulsions of liquids, gases and solids) as may be used in various operations to drill boreholes into the earth. Classifications of drilling fluids may utilize one or more types of classification schemes. For example, consider water-based mud (WBM), oil-based mud (OBM), nonaqueous-based mud (NQBM), gaseous-based mud (e.g., pneumatic, etc.) (GBM), etc.

[0093] As an example, drilling fluid may be lost to a formation and/or reservoir fluid may enter drilling fluid. Hence, one or more functions of drilling fluid may be compromised by changes to drilling fluid. For example, if density of drilling fluid is altered by introduction of reservoir fluid, the drilling fluid may diminish in its ability to transport cuttings to surface. As a consequence, cuttings may build up within an annulus between a drillstring and a borehole wall or cased wellbore, which may increase risk of sticking (e.g., stuck pipe). To address changes to drilling fluid, one or more actions may be taken, for example, consider adding one or more components to the drilling fluid, adding additional drilling fluid, etc.

[0094] As to lost circulation or circulation loss, these terms may refer to loss of drilling fluid to a formation, for example, caused when the hydrostatic head pressure of the column of drilling fluid exceeds the formation pressure. This loss of fluid may be loosely classified as seepage losses, partial losses, or catastrophic losses, each of which may be handled differently depending on the risk to equipment, materials, borehole quality, characteristics of drilling fluid, personnel, etc.

[0095] As to influx of formation fluid (e.g., reservoir fluid, etc.), it may include an event known as a kick. A kick may be defined as a flow of formation fluid into a bore during drilling operations. A kick may be physically caused by the pressure in the bore being less than that of the formation fluid, thus causing flow. This condition of lower bore pressure than formation pressure may be caused in various ways. For example, if mud weight is too low, then hydrostatic pressure exerted on a formation by the fluid column may be insufficient to hold the formation fluid in the formation. This may happen if the mud density is suddenly lightened or is not to specification to begin with, or if a drilled formation has a higher pressure than expected. This type of kick might be called an underbalanced kick. As another example, consider a kick that may occur if dynamic and transient fluid pressure effects (e.g., due to motion of the drillstring or casing), effectively lower the pressure in a bore below that of the formation. Such a type of kick may be referred to as an induced kick.

[0096] Additional phenomena that may occur during drilling operations include swab and surge. As to swab, it may involve a reduction in pressure in a bore by moving pipe, wireline tools or where rubber-cupped seals up the bore. If the pressure is reduced sufficiently, reservoir fluid may flow into the bore and towards surface. Swabbing tends to be detrimental in drilling operations as it may lead to kick and borewall stability problems. As to surge, consider an example where a drillstring is being tripped-out (e.g., pulled out of hole (POOH)) where upward movement of the drillstring causes friction between the drillstring and drilling fluid. In surge, pressure may decrease in a bore due to the surge effect; noting that the opposite effect may happen when a drillstring is tripping-in (e.g., running in hole (RIH)), as downward movement may cause a pressure increase (e.g., a swab effect).

[0097] As explained, a drillstring may include a mud motor that is rotationally driven by flow of drilling fluid. In such a mode of drilling, the characteristics of drilling fluid may impact mud motor performance. For example, density (e.g., mud weight) may impact how much energy the mud motor may deliver to a drill bit for a given drilling fluid flow rate.

[0098] As to a stuck pipe or risk of sticking event, as explained, one or more actions may be taken. For example, consider addition of acid as a remedial action to address the stuck pipe event or to reduce the risk of a sticking event. In such an example, a number of barrels of acid may be added to drilling fluid that is circulated downhole to an annular region between a drilling string and a bore wall in an effort to "dissolve" material that is causing sticking or a risk of sticking. While addition of acid is mentioned, it may be an action within a tiered series of actions that may be taken, where, for example, each action may have associated benefits and detriments. As to detriments, these may include non-productive time (NPT), cost, further remedial actions (e.g., impact of acid on one or more additives in drilling fluid), etc. Hence, where an event occurs or a risk of an event is heightened, in an effort to maintain adherence to a plan, one or more actions may be implemented in a strategic manner to resolve the event or otherwise reduce the risk.

[0099] FIG. **3** shows an example of a wellsite system **300**, specifically, FIG. **3** shows the wellsite system **300** in an approximate side view and an approximate plan view along with a block diagram of a system **370**.

[0100] In the example of FIG. **3**, the wellsite system **300** may include a cabin **310**, a rotary table **322**, drawworks **324**, a mast **326** (e.g., optionally carrying a top drive, etc.), mud tanks **330** (e.g., with one or more pumps, one or more shakers, etc.), one or more pump buildings **340**, a boiler building **342**, an HPU building **344** (e.g., with a rig fuel tank, etc.), a combination building **348** (e.g., with one or more generators, etc.), pipe tubs **362**, a catwalk **364**, a flare **368**, etc. Such equipment may include one or more associated functions and/or one or more associated operational risks, which may be risks as to time, resources, and/or humans.

[0101] As shown in the example of FIG. **3**, the wellsite system **300** may include a system **370** that includes one or more processors **372**, memory **374** operatively coupled to at least one of the one or more processors **372**, instructions **376** that may be, for example, stored in the memory **374**, and one or more interfaces **378**. As an example, the system **370** may include one or more processor-readable media that include processor-executable instructions executable by at least one of the one or more processors **372** to cause the system **370** to control one or more aspects of the wellsite system **300**. In such an example, the memory **374** may be or include the one or more processor-readable media where the processor-executable instructions may be or include instructions. As an example, a processor-readable medium may be a computer-readable storage medium that is not a signal and that is not a carrier wave.

[0102] FIG. **3** also shows a battery **380** that may be operatively coupled to the system **370**, for example, to power the system **370**. As an example, the battery **380** may be a back-up battery that operates when another power supply is unavailable for powering the system **370**. As an example, the battery **380** may be operatively coupled to a network, which may be a cloud network. As an example, the battery **380** may include smart battery circuitry and may be operatively coupled to one or more pieces of equipment via a SMBus or other type of bus.

[0103] In the example of FIG. **3**, services **390** are shown as being available, for example, via a cloud platform. Such services may include data services **392**, query services **394** and drilling services **396**. As an example, the services **390** may be part of a system such as the system **100** of FIG. **1** (e.g., consider planning services and/or operational services). As an example, the services **390** may include one or more services for directional drilling (e.g., consider a computational framework that may provide for one or more services that utilize real-time data to estimate one or more parameters, etc.).

[0104] As an example, the system **370** may be utilized to generate one or more rate of penetration drilling parameter values, which may, for example, be utilized to control one or more drilling operations.

[0105] FIG. **4** shows an example of a system **400** that includes offsite equipment **401** (e.g., remote) and onsite equipment **402** (e.g., local). As shown, the offsite equipment **401** may include a drill operations framework **410**, a drill planning framework **420** and a database **430** and the onsite equipment **402** may include a controller **440** that may receive real-time data and output recommendations such as control instructions to control onsite equipment. In such an example, the drill operations framework **410** may provide for steering sheets, execution parameters, etc., and the drill planning framework **420** may provide for evaluation of steering responses and statistics. As shown, the controller **440** may output information to the drill operations framework **410** and receive information from the drill planning framework **420**. The system **400** may include plan generation features for real-time plan generation during drilling operations execution phase and/or plan generation during a planning phase. The system **400** may be utilized for one or more types of drilling (e.g., rotary, mud motor, RSS, ABSS, etc.). The system **400** may operate loops, which may include at least one real-time loop that provides for control of equipment to perform drilling operations.

[0106] A system such as the system **400** may utilize various functions and constraints for generation of plans, which may provide for single or multiple target aiming. As explained, a plan may be generated that aims to provide for drilling operations for a multiwell structure. As explained, a plan may be a digital plan that may be utilized to instruct one or more controllers such as, for example, an autodriller controller, which may control one or more pieces of equipment (e.g., a drawworks, a top drive, one or more drilling fluid pumps, etc.). As an example, an autodriller may control energy delivered via one or more pieces of equipment to a drill bit where the drill bit crushes and/or cuts rock to extend a borehole. As an example, an autodriller may be controlled in an effort that aims to minimize or otherwise reduce mechanical specific energy (MSE) and to maximize or otherwise increase rate of penetration (ROP).

[0107] As an example, control of one or more field operations at a wellsite may be facilitated through use of a framework that may provide one or more actions for addressing an event, a risk of an event, etc. For example, consider a framework that may allow an engineer to interact with the framework via one or more graphical user interfaces (GUIs) and/or one or more other types of user interfaces (UIs) (e.g., microphone, QR code, etc.).

[0108] As an example, the system **400** may include one or more global drilling mechanics models (GDMMs). As an example, a GDMM may be generated that may be used for one or more purposes, which may include, for example, one or more of drill planning, offset well analysis, drilling performance evaluation, drilling monitoring, drilling optimization, etc. As an example, a GDMM may be applied for well construction engineering. As an example, a GDMM may provide for receipt of drilling mechanics data, organization and storage of data, search and retrieval data, and various computational components that may utilize data and/or prepare data for one or more workflows (e.g., consider data subsets, etc., tailored for one or more particular workflows). As an example, a GDMM may be utilized for one or more purposes, which may include one or more of drilling, modeling building, seismic interpretation, etc. As explained, a GDMM may be integrated into one or more control loops that may provide for improved field operations, which may include control of one or more field operations, which may occur at one or more levels of automation.

[0109] As an example, one or more techniques, technologies, etc., may be applied to data such as, for example, natural language processing (NLP), large language model (LLM) processing, image analysis (e.g., one or more image-tailored machine learning models), etc.

[0110] FIG. **5** shows an example of a system **500** that may provide for generation and/or utilization of a GDMM. As shown, the system **500** may provide for receipt of data from various sources, which may include one or more of a drilling record system (DRS, e.g., recording bit performance), the QTRAC platform (SLB, Houston, Texas; e.g., recording BHA and directional drilling performance), the Performance Took Kit (PTK) platform (SLB, Houston, Texas; e.g., drilling data ingesting and interpretation), the DRILLOPS framework, one or more tool memory dumps (e.g., tool memory downloads, etc.), etc. As shown, data may be organized in the form of well construction (WC) data, which may provide for organization with respect to tool equipment (BHA, etc.), a trajectory, wellbore geometry, entities (e.g., operator, contractor, etc.), etc. The organized data may be utilized to generate and/or support operation of a GDMM. As an example, the data may be contextualized. For example, contextual information may be determined via one or more types of information and/or one or more techniques (e.g., consider data-driven, natural language processing (NLP), data extraction, etc.).

[0111] As shown, a GDMM may be or include a type of information and knowledge base that leverage a well construction data foundation (WCDF), domain knowledge, and one or more physics-based models and/or data-based models (e.g., consider machine learning types of models).

[0112] As an example, data may be accessed and organized as to formation tops and drilling indices, well construction data (e.g., BHA, trajectory, wellbore geometry, operator, contractor, etc.), operation parameters, drilling dynamics and/or mechanics, performance indicators, IAR digital results (e.g., drilling simulation results), etc. As an example, one or more computational components may provide for implementation of one or more types of models (e.g., physics-based and/or data-driven models). As an example, such computational components may provide for: formation top interpolation; offset data selection given geo-location, BHA, etc.; depth and feature alignment using offset data for given geo-location; critical drilling depth identification; the drill bit optimization system (DBOS, SLB, Houston, Texas) to the IDEAS framework (SLB, Houston, Texas) rock file mapping; calibrated i-DRILL framework formation models (SLB, Houston, Texas), etc.

[0113] The DBOS is a system that may be implemented in combination with the TECHLOG framework (SLB, Houston, Texas), for example, to provide well logs, formation tops, mud logs, rock mechanics, core analysis, bit records, real-time drilling parameters, dull grading condition, and survey data from more than 30,000 wells around the world. Hence, the DBOS may provide global data for use in generating and/or implementing a GDMM.

[0114] As to the IDEAS framework, it is an Integrated Drilling dynamics Engineering Analysis System that may provide 4D simulations of an entire drillstring and wellbore geometry to help ensure accurate modeling for drilling applications. The IDEAS framework provides for assessment of interactions in a virtual environment, which may be tailored to an actual environment and/or realizations (e.g., statistical, etc.) thereof. As an example, the IDEAS framework may be implemented in a manner that may allow for customization of material and/or design in real time. Such features may be utilized to help predict bit performance while reducing trial-and-error field tests, which may provide for achieving desired results on a first run.

[0115] As to the i-DRILL framework, it may provide for analyzing drilling assembly dynamics through varying formations to assist in optimizing drilling performance. As an example, an i-DRILL framework model may provide for simulating behavior of a bit, as well as components of a BHA and a drillstring, in a virtual drilling environment, which may be based on an actual environment and/or realizations thereof. As an example, a workflow may include simulating drilling operation to enable evaluation of a root cause of inefficient and/or damaging BHA behavior. As an example, results may provide for evaluating a range of options to reduce harmful vibrations, thus increasing equipment life and minimizing failures, increasing ROP, improving hole condition and directional control, and decreasing overall drilling resource utilization. By providing analyses of real-world simulations of downhole drilling properties, the i-DRILL framework may help to reduce unnecessary trips to change BHAs, identify true technical limitations without risking lost time, predict different BHA behaviors, identify weak points in a drillstring and equipment, and minimize vibrations and stick/slip. As an example, the i-DRILL framework may provide for analyses as to causes of vibrations, failures, and inefficiencies; determinations as to an appropriate bit and cutter design as well as weight on bit (WOB) and RPM ratios for maximum ROP; investigations as to troubled BHA behavior; and performance of post-well follow-ups, which may help to determine efficiency of i-DRILL framework design recommendations.

[0116] FIG. **6** shows an example of a GDMM **600** that may generate output as to run context, run summaries, data labels (e.g., quality control (QC) marking for workflows, labels for answer products, etc.), run time series, run depth series, trained and/or calibrated models.

[0117] As to run context, consider a GDMM that may operate using information about a well, a drilling system and a well construction process. For example, as to a well, consider location, entity, well global ID (e.g., wellGUID), trajectory, wellbore geometry, formation tops, drilling indices, etc. As to a drilling system, consider bit, BHA, drillstring, rig, surface equipment, control systems, mud type, properties, etc. As to a well construction process, consider auto driller and/or manual control, use of one or more platforms (e.g., OPTIWELL framework, SLB, Houston, Texas), live performance tools, remote operation (RO), directional drilling advisors, a planning framework (e.g., the DRILLPLAN framework), a drilling operations framework (e.g., the DRILLOPS framework), directional drilling indexes (DDIs), optimum operation window (OOW), offset well analysis (OWA), the i-DRILL framework, IAR, etc.

[0118] As to the OPTIWELL framework, it may provide for construction performance service monitoring and analyses of processes and downhole conditions in real time. Such a framework may facilitate improvement in overall safety and efficiency of a well construction process. As an example, the OPTIWELL framework may provide for monitoring as to downhole hazards, which may provide for identifying inefficiencies and/or mitigating HSE risks and, for example, invisible lost time (ILT) and non-productive time (NPT).

[0119] As to run time series and run depth series, consider a GDMM that may operate to utilized and/or to output various surface channels (e.g., HKLD, SPPA, STOR, WOB_SP, etc.), real-time D-points (e.g., inclination, azimuth, lateral vibration, etc.), computed rig states, computed drilling states (e.g., using automated state detection, etc.), real-time interpretation channels (e.g., motor efficiency, bit aggressiveness, automated drilling controller stability, mud loss, etc.), tool channels (e.g., via tool memory, which may be accessed at surface via a memory dump, etc.), interpretation channels from dump files (e.g., micro stall, DLS, tortuosity, etc.), etc.

[0120] As to run summaries, consider a GDMM that may operate to utilize and/or to output time series of channels using statistics (e.g., by rig state), accumulated values, etc.; depth series of channels using statistics (e.g., for tool yield, tortuosity, hard stringer (HS), interbed (IB), etc.), accumulative values (e.g., tortuosity, sliding ratio, sliding efficiency, etc.); performance summaries as to metrics such as footage, footage per day, ROP on bottom, on bottom time versus off bottom time, BRT, average connection time, etc.; tool health and/or damage status such as bit dull, motor degradation, RSS damage, LWD damage, MWD damage, special events, lost circulation, twist-off (e.g., of one or more components of a drillstring, etc.), washout, stuck pipe, top drive stall, downhole motor stall, etc.

[0121] As to formation tops (e.g., well tops or formation boundaries) and associated drilling indexes, these may be utilized for generating a GDMM. For example, consider a workflow that includes building a global grid with formation top information. In such an example, given latitude and longitude coordinates and total vertical depth (TVD) or measured depth (e.g., via a survey), a generated GDMM may provide formation name as output. As an example, in addition to formation tops, a workflow may include assigning drilling indices to formations. For example, consider accessing data in a platform such as, for example, the DBOS platform. As an example, formation top information and drilling indices may be accessed via a well construction data foundation (WCDF) framework, which may include formation tops from one or more data sources (e.g., public survey data, private survey data, etc.). As an example, information may be accessed from the DBOS platform that includes formation tops and depths, rock mixtures, formation drilling indices (e.g., UCS, abrasion, impact, etc.), and surveys.

[0122] As explained, the DBOS includes data from around the world. As an example, a workflow may include compiling over 20,000 DBOS datasets, including formation tops and depths, rock mixtures, UCS, impact indices, abrasion indices, etc.

[0123] As an example, mechanical specific energy (MSE) may be utilized to build and/or implement a GDMM. MSE may be a measure of drilling efficiency and may be defined as an amount of energy required to remove a unit volume of rock (e.g., a unit volume of a formation). For optimal drilling efficiency, an objective may be to minimize MSE and to maximize rate of penetration (ROP). To control MSE, drilling operations may aim to control the weight on bit (WOB), torque, ROP, and drill bit revolutions per minute (RPM). As an example, MSE may encompass both surface and downhole mechanical specific energy where, for example, downhole MSE may be referred to as DMSE.

[0124] As an example, drilling indices may be constructed using drilling data together with data models and/or physics models. Some examples of such indices may include shock and vibration (S&V) indices from tool dumps, DMSE from surface/downhole channels, DLS normalized by RSS tool steering ratio in the curve from continuous D&I, tortuosity from continuous D&I, borehole over gauge from caliper (e.g., using the ECOSCOPE tool, SLB, Houston, Texas), friction coefficients along wellbore from T&D inversion, etc. As an example, to generate such indices, data processing, enrichment computations and/or modeling may be performed. For example, consider one or more of surface and dump data synchronization, data enrichment (e.g., rig state, DWOB, DTOR, differential pressure), tool memory data depth gating, and domain computation (e.g., for vibration indices, DLS and tortuosity from continuous D&I, motor efficiency, degradation, power, torque, MSE, DMSE, torque loss, bit aggressiveness, formation stiffness, hard stringers (HSs) and interbeds (IBs) flags and grades, friction coefficients along wellbore from T&D inversion, i-DRILL calibrated formation model, etc.

[0125] FIG. **7** shows an example GUI **700** of log data, including density caliper average (DCAV), ultrasonic caliper average (UCAV), caliper image (CAL_IMG), gamma ray average (GRMA), American Petroleum Institute gamma ray units (gAPI), and inclination (INC_GX_ECO) in degrees with respect to depth. As mentioned, borehole over gauge from caliper may be utilized, which may be acquired using a tool such as, for example, the ECOSCOPE tool.

[0126] FIG. **8** shows an example of a workflow **800** for various formation type determinations, which may include, for example, determinations as to hard stringer (HS) and interbed (IB) types of formations. The workflow **800** may include receiving data such as data presented in a plot **810** of DMSE versus measured depth. For example, data as to DMSE with respect to depth may be assessed as to drilling zones where, for example, drilling zones may be partitioned based at least in part on DMSE.

[0127] As shown in FIG. **8**, the workflow **800** may utilize a model **820** to make drilling zone type partitions with respect to measured depth. For example, consider a decision tree type of model where decisions may be made on the basis of comparing data to one or more criteria. In the example of FIG. **8**, the workflow **800** may be implemented using DMSE versus depth data (e.g., true vertical depth and/or measured depth) to make determinations as to drilling zone types, which may include, for example, one or more of normal formation (NF), interbedded formation (IB), medium hard stringer (MHS), medium hard interbed (MHI), hard stringer (HS), hard interbed (HI), and high shock and vibration interval (SV). As to the workflow **800**, one or more of the high shock and vibration intervals (SVs) may be determined for particular types of drilling zones (e.g., NF, IB, and MHS) on the basis of values for stick-slip index (SSI_SVR), axial shock and vibration index (AXL_SVR) and/or lateral shock and vibration index (LAT_SVR). For example, decision logic may include comparing to levels such as severe and high, where severe is a higher level (e.g., worse or more detrimental) than high. As shown, such decision logic may include determining that a drilling zone is a high shock and vibration zone (e.g., interval) if SSI_SVR>=server or AXL_SVR>=high or LAT_SVR>=high.

[0128] As to hard stringers, drilling hard stringers that may be erratically distributed in a subsurface environment may be challenging, particularly where the underlying subsurface environment tends to be relatively soft formation with such erratically distributed hard stringers. In such an example, an unforeseen change of the drilled formation from soft to hard and dense rock may cause impact damage to a bit, deflect a BHA, result in high bending loads, increase vibration, and cause wear/tear on BHA components. If not properly managed, such unforeseen changes may lead to an increase in NPT and maintenance costs. Further, a deflection caused by a stringer away from a planned trajectory that may be detected relatively late may result in excessively high local doglegs and demand time-consuming adjustments (e.g., through reaming, etc.), which may introduce ILT.

[0129] FIG. **9** shows an example of a GUI **900** that includes various data and risk assessments with respect to depth for various types of formation zones, along with formation types. As shown, the GUI **900** may include horizontal indicators as to types of formation zones, data as to ROP, STOR, and MSE (MSE_DDI), along with risk assessments as to lateral shock and vibration risk, axial shock and vibration risk, and torsional shock and vibration risk, which may be coded at various levels (e.g., acceptable or no risk, low risk, medium risk and high risk).

[0130] As to DLS computations and normalization, a workflow may include normalizing DLS by steering ratio for RSS tools to indicate tool directional capability. For example, consider a workflow that may include accessing various DLS data along with azimuth and inclination data, which may be from various sources (e.g., static surveys, etc.). In particular, one or more types of inclination data may be utilized (e.g., from a definitive survey, from a static surface, and from a TRULINK dynamic, directional survey (DDS), SLB, Houston, Texas). As an example, the TRULINK DDS may provide for telemetry uphole at approximately **20** bps where advanced drilling dynamics include three-axis shock and vibration and turbine power. As an example, geological accuracy may be refined using one or more other types of data such as, for example, EM data and/or gamma ray data, optionally in combination with continuous six-axis directional and inclination sensors. As an example, real-time data may provide for a drilling a borehole with a tighter, more accurate curve in complex 3D profiles where sustaining directional control (e.g., inclination and azimuth) is required, while minimizing tortuosity that may accompany drilling of a dogleg.

[0131] FIG. **10** shows an example of a method **1000** that provides for borehole friction inversion using drilling data and torque and drag (T&D) computations. As shown, the method **1000** may include an input block **1010** for receiving BHA data, trajectory data, mud weight data, and one or more other types of surface data. In such an example, an inversion block **1020** may provide for inverting for borehole friction using T&D computations for each stand of drill pipe during an off-bottom rotation state and/or during tripping out of hole (e.g., pulling out of hole, POOH). In such an example, the method **1000** may include an output block **1030** for outputting borehole friction (e.g., BH friction) values with respect to measured depth (MD) (see also, e.g., FIG. **21** and use of BH friction as to assessment for borehole cleaning (BHC)).

[0132] As explained, a GDMM may be generated and/or implemented using various types of data. For example, consider planning of drilling operations for a well, which may be referred to as a target well, which is specified to be drilled at a location, which may be within a field encompassed by a GDMM.

[0133] FIG. **11** shows an example of a GUI **1100** that includes various examples of data that may be organized to form a model for a well where the model may be considered a GDMM or based on a GDMM. As an example, information may include BHA information and various types of information organized with respect to depth (e.g., TVD and/or MD). As an example, consider information organized with respect to depth as to formation tops, rock mixtures, UCS, abrasion index, impact index, IDEAS analysis and i-DRILL rock files, over gauge, borehole friction, normalized DLS, lateral vibration risk, stick-slip risk, DMSE, hard string and grade, and interbed and grade; noting that one or more types of additional and/or alternative information may be included.

[0134] As an example, a framework may include a meta table that may organize information as to GDMM data. For example, consider a meta table that may provide for searching and/or filtering data, which may be via one or more of context, performance, failure and risks, data labels to locate the particular information to retrieve time and depth series, data models, etc. As an example, meta data may also be utilized to drive quick decision making such as bit recommendation, motor recommendation, etc. As explained, a run may be contextualized and/or summarized. As explained, labels may be generated for one or more purposes (e.g., quality control, machine learning, etc.). As an example, a framework may provide for generation of a particular model suitable for drilling operations at a particular location, along with identification of data that may be germane to such drilling operations. As explained, a framework may provide for accessing data for over ten thousand wells throughout portions of the world such that the framework may be suitable for global use.

[0135] In a particular example, over 11,000 sets of synchronized downhole MWD measurements and surface data with time and depth series, drilling indices in time and depth series (bit indices, motor indices, S&V indices, drilling parameters, etc.), were utilized; noting that such data may be supplemented responsive to new drilling activities. Data may include 25,000 sets of surface data time series and depth series, over five million sets of DRS data, over 500,000 sets of QTRAC data (e.g., enhanced with surface parameter statistics, S&V statistics, bit and motor indices, PD directional indices, etc.), over 20,000 sets of DBOS data (e.g., formation top and depth, rock mixtures, formation drilling indices, etc.), and over 80,000 trajectories (e.g., runs having an associated trajectory).

[0136] As to GDMM formation top interpolation, formation tops and depths may be utilized as a fundamental reference in a GDMM. As an example, a framework may utilize formation tops and depths to help organize other indices. As an example, a framework may implement a localized formation top interpolation process that may utilize IHS data (e.g., survey data provided by IHS Markit). As an example, a framework may generate formation top interpolations for a number of relevant basins using formation tops from one or more of DBOS, IHS and one or more other privately collected data sources.

[0137] As an example, a framework may provide for selection of suitable offset data, which may be utilized for one or more purposes such as, for example, to drive planning, monitoring and post job interpretation workflows. As an example, a framework may provide for manual and/or automated selection of offset data, for example, based on one or more of geo location distance, trajectory, BHA, bit similarity, depth in and depth out, bit dull grading, entities (e.g., operators), etc.

[0138] FIG. **12** shows an example of a method **1200** for offset data alignment using formation tops as a reference. As shown, the method **1200** may include implementing a filtering process **1210** for filtering offset data based on one or more criteria using well metadata such as BHA, well type, size, etc. As an example, the method **1200** may include implementing a preliminary assignment process using one or more formation top interpolation functions to assign formation tops for offsets, for example, if one or more of such assignments may not be present.

[0139] As shown in the example of FIG. **12**, the method **1200** may include implementing an alignment process **1220** for aligning drilling data using formation tops where, for example: statistics of formation top depth are obtained using offset data, followed by picking median or average depth as representing depth for the offset wells; stretching or compressing section length based on representing formation top depth for offset wells to align drilling indices and indicators; and generating statistical data of such indices or indicators to represent the offset wells. As an example, one or more other techniques may be utilized, additionally or alternatively, to align data (e.g., via one or more of TVD, trajectory, (kick off, etc.), data channel features, etc.). As an example, a machine learning-based approach may be implemented to align offset data and/or for formation top and depth organization and/or analysis.

[0140] As shown in the example of FIG. **12**, the method **1200** may include outputting process **1230** for outputting one or more representative wells where formation tops are indicated with respect to depth (e.g., total vertical depth (TVD)). As an example, information from offset wells may be represented at an indicator of formation tops with respect to depth for a target well.

[0141] FIG. **13** shows an example of a GUI **1300** that includes a map of a particular field, specifically the Permian Basin. As an example, a framework may be utilized to facilitate drilling operations for a well in the Permian Basin. As an example, a GDMM may include information as to wells with DBOS formation tops and drilling indices (open circles) and wells with enriched drilling data or computed domain indices (black filled circles). As an example, a model may be applied for one or more purposes, which may include one or more of: formation tops interpolation during planning; critical depth for IDEAS analysis and rock file mapping; performance estimation based on offsets; optimum operation window (OOW) planning; real-time formation top or drilling condition expectations; real-time bit damage estimation and POOH decision making; real-time DLS monitoring and parameter planning; and real-time hole condition monitoring.

[0142] As an example, a framework may provide for identifying one or more issues associated with a planned trajectory for a target well at a location. In such an example, one or more issues may be addressed via performing one or more simulations. For example, consider a method that may include generating aligned offset well data where one or more actions as to a target well may be focused. For example, consider driving one or more simulators based on one or more criteria as may be utilized to assess aligned offset well data. In such an example, where shock and vibration risk may be heightened, one or more criteria may be utilized to trigger one or more types of simulations to generate simulation results for assessment of the shock and vibration risk. In such an example, simulation runs may be focused on one or more portions of a planned borehole trajectory, etc., which may provide for expedited and/or improved drilling (e.g., compared to a human-based assessment that may be overly comprehensive and demand considerable computational resources for performing simulations).

[0143] As an example, a framework may provide for detailed bit assessments, which may be integrated with one or more types of simulators, if warranted. As an example, if a real-time MSE is elevated during drilling of a borehole using a drillstring with a drill bit in comparison to aligned offset well MSE data, that may be an indication that the drill bit has experienced excessive wear, which may be assessed via a simulation, etc., for example, to determine how much bit life the drill bit has and whether the bit life is sufficient to complete drilling of a current section of the borehole. In such an example, if the remaining bit life is insufficient, one or more drilling parameters may be adjusted such as, for example, adjusting one or more drilling parameters of an auto driller, and/or a tripping out of the drill string may be planned (e.g., scheduled sooner rather than later, etc.).

[0144] As explained, a GDMM framework may be employed in combination with a planning framework and/or a drilling operations framework (e.g., the DRILLPLAN framework, the DRILLOPS framework, etc.). As explained, a GDMM framework may be implemented in real-time during drilling where, for example, output thereof may be utilized by one or more controllers to control performance of drilling.

[0145] As an example, a GDMM framework may include one or more features for restricting data, which may include restricting access, restricting names, etc. As an example, a GDMM framework may automatically control access, search, filtering, etc., based on one or more criteria, which may include location of a well, location of a request, location of data, etc.

[0146] FIG. **14** shows an example of a method **1400** that may provide for formation top interpolation. As shown, the method **1400** may include a filtering process **1410** that may be implemented using a GDMM, a formation top generation process **1420** that may involve kriging and/or one or more other techniques, and an output process **1430** that may provide for outputting formation tops with respect to depth for a target well, for example, where formation tops may be projected to a target well TVD and/or measured depth, which may operate dynamically if trajectory design is varied.

[0147] As explained, a framework may provide for filtering and application of one or more formation top interpolation functions to assign formation tops for offset wells if such assignments do not exist and/or may be of questionable accuracy (e.g., via a quality control assessment, etc.). As shown, the method **1400** may include performing formation top interpolation whereby formation tops may be projected to a target well as to measured depth along a planned trajectory to be drilled for the target well, which may allow for varying the planned trajectory prior to and/or after drilling has commenced for the target well.

[0148] FIG. **15** shows an example of a GUI **1500** that includes various information for a target well. For example, the GUI **1500** may be rendered via execution of a framework where depth and rock file selection may be performed automatically responsive to input for a target well (e.g., one or more target well criteria, etc.). As shown, a framework may provide for automated selection of depth of analysis thereof based on GDMM determined offsets, where information as to one or more of hard stringers, interbeds, high S&V, DLS challenges, may be presented. As explained, a framework may provide for automated selection of rock filles, which may provide, for example, one or more i-DRILL calibrated formation models, one or more DBOS rock file mappings, etc.

[0149] FIG. **16** shows an example of a method **1600** that provides for offset well performance estimation using a GDMM framework. As shown, the method **1600** may include various processes, such as, for example, a filtering process **1610** for filtering data for various wells with respect to a target well, a drilling index and/or performance index process **1620** for generating one or more indexes with respect to depth, and an output process **1630** for outputting one or more indexes for a target well with respect to depth (e.g., TVD and/or measured depth).

[0150] As explained, a framework may provide for filtering and application of one or more formation top interpolation functions to assign formation tops for offset wells if such assignments do not exist and/or may be of questionable accuracy (e.g., via a quality control assessment, etc.). As shown, the method **1600** may provide for offset data being manually and/or automatically selected and aligned based on formation tops. In such an example, drilling indices and performance indicators may be obtained. For example, specific performance estimation such as ROP and DLS may utilize offset data, one or more machine learning models, one or more physics-based models, one or more hybrid models, etc. As shown in the example of FIG. **16**, final performance may use formation tops and depth as references.

[0151] FIG. **17** shows an example of a process **1700** that may provide for optimal operation window (OOW) generation using a GDMM framework. As shown, the process **1700** may include a reception block **1710** for receiving offset well data for a targeted application, an OOW generation block **1720** that utilizes the offset well data and one or more types of models (e.g., physics-based, data-driven, planning logistical, etc.), and an OOW output block **1730** that may output a generated OOW as organized data, optionally in graphical form as may be rendered to a display. In such an approach, a GDMM block **1740** may provide for accessing and/or supplementing formation tops, drilling data, drilling indices, performance indicators, contexts, etc., which may thereby enrich offset well data for one or more other projects (e.g., runs, etc.). As shown, the process **1700** may be iterative in that the GDMM block **1740** may provide for GDMM enhancement responsive to OOW generation and/or implementation (e.g., field implementation at one or more sites, etc.).

[0152] FIG. **18** shows an example of a GUI **1800** that provides for real-time formation top and/or drilling condition expectations using a GDMM framework. As shown, during execution of such a GDMM framework, formation tops, drilling indices, etc., may be generated and utilized to indicate one or more types of expected drilling conditions where, in response, one or more drilling operations may be adjusted accordingly. In such an example, a framework may be operatively coupled to one or more controllers, which may be part of a rig control system, etc. For example, consider one or more automated or semi-automated responses to an expected condition whereby a control command may be issued to a controller to adjust a drilling operation to handle the expected condition (e.g., a risk, a benefit, etc.). As to a risk, consider identification of an expected hard stringer at a particular measured depth whereby one or more drilling parameters may be suitable adjusted to drill through the hard stringer and/or to drill around the hard stringer, if practical.

[0153] As an example, the GUI **1800** may provide for rendering an alert track that may include various types of alerts. As shown, an alert for an expected hard stringer (HS) may be rendered along with a numerical value that may indicate a distance, which may, for example, be dynamically updated in real-time. As an example, an alert may be provided with one or more adjustable parameters such as, for example, distance to a hazard and/or other condition, geology, etc. In such an approach, a user may tailor one or more parameters to provide for earlier or later alerts, which may depend on type of issue and/or how much time it may take to control operations to address the issue.

[0154] FIG. **19** shows an example of a GUI **1900** that provides for real-time bit damage estimation and tripping out decision making using a GDMM framework. As shown in the example of FIG. **19**, bit damage may be estimated based on offset well bit damage indices (see, e.g., bit index) and, for example, a tripping out (e.g., POOH) decision may be made based on bit damage severity and expected damage to finish a section currently being drilled.

[0155] As an example, a bit index may take into account one or more parameters such as, for example, one or more of WOB, rotary speed (RPM), hydraulic horsepower (HHP), ROP, torque, bit footage, dull bit condition, etc. As an example, a group of parameters may be utilized to compute bit index values, for example, at desired intervals. As an example, a GUI may present bit index and/or MSE (e.g., DMSE, etc.), noting that bit index and MSE may be related. As an example, a bit index may be computed as a metric that is indicative of bit wear and/or remaining bit life. As an example, a bit index may be a bit damage index that increases in value as bit damage increases.

[0156] As an example, a bit index may be provided that is an estimated pre-job bit index and may be provided that is an estimated real-time bit index. As shown, the real-time bit index may increase more than the pre-job bit index, which may cause issuance of an instruction such as an alert in an alert track that indicates depth and/or time to POOH for servicing a bit of a drillstring. In such an example, a framework may provide for re-planning and/or re-estimating one or more types of information germane to a job, which may include re-planning for one or more sections such as, for example, starting a new section earlier than originally planned due to tripping out of the drillstring earlier than originally planned.

[0157] FIG. **20** shows an example of a GUI **2000** that provides for real-time DLS monitoring using a GDMM framework. As shown, real-time DLS monitoring may be performed using a GDMM framework for drilling of a target well where, for example, if normalized DLS is below a certain threshold as a criterion, the GDMM framework may call for adjusting one or more drilling parameters and/or checking tool health status.

[0158] As an example, a metric such as normalized DLS may be provided that is an estimated pre-job normalized DLS and may be provided that is an estimated real-time normalized DLS. As shown, the real-time normalized DLS may be less than the pre-job normalized DLS at a particular depth or depths, which may cause issuance of an instruction such as an alert in an alert track that indicates depth and/or time to adjust one or more operational parameters for achieving a desired (e.g., planned) DLS. In such an example, a framework may provide for re-planning and/or re-estimating one or more types of information germane to a job, which may include re-planning for one or more sections.

[0159] FIG. **21** shows an example of a GUI **2100** that provides for real-time borehole friction monitoring using a GDMM framework. As shown, real-time borehole friction monitoring may include issuing one or more commands for checking ECD, improving borehole cleaning, adding borehole lubricant, etc.

[0160] As an example, a metric such as borehole friction may be provided that is an estimated pre-job borehole friction and may be provided that is an estimated real-time borehole friction. As shown, the real-time borehole friction may be greater than the pre-job borehole friction at a particular depth or depths, which may cause issuance of an instruction such as an alert in an alert track that indicates depth and/or time to assess borehole conditions such as, for example, to determine whether one or more borehole cleaning operations are warranted for achieving a desired (e.g., planned) amount of borehole friction. In such an example, a framework may provide for re-planning and/or re-estimating one or more types of information germane to a job, which may include re-planning for one or more sections.

[0161] As an example, a GDMM framework may be implemented using one or more computational resources, platforms, etc., which may include local and/or remote resources. As an example, a GDMM framework may implement one or more network architectures where, for example, one or more application programming interfaces (APIs) may be utilized. For example, consider search and retrieval functionality that may leverage one or more APIs. As an example, a GDMM framework may be implemented within a framework and/or operatively coupled to a framework. For example, consider implementation in a manner coordinated with the TECHLOG framework and the DRILLOPS framework.

[0162] FIG. **22** shows an example of a method **2200** and an example of a system **2290**. As shown, the method **2200** may include a reception block **2210** for receiving data for drilling of a borehole in a subsurface environment according to a borehole trajectory for a target well at a field location in a basin; an access block **2220** for accessing offset well data for wells in the basin using at least a portion of the data, where the offset well data include identified formation tops with respect to well depth for a number of formations within the basin and mechanical specific energy drilling data with respect to well depth; an alignment block **2230** for aligning the offset well data using the formation tops to generate aligned offset well data that specify one or more drilling zones based on the mechanical specific energy drilling data; and a performance block **2240** for performing the drilling based at least in part on the aligned offset well data.

[0163] FIG. **22** also shows various computer-readable media (CRM) blocks **2211**, **2221**, **2231**, and **2241**. Such blocks may include instructions that are executable by one or more processors, which may be one or more processors of a computational framework, a system, a computer, etc. A computer-readable medium may be a computer-readable storage medium that is not a signal, not a carrier wave and that is non-transitory. For example, a computer-readable medium may be a physical memory component that may store information in a digital format.

[0164] In the example of FIG. **22**, a system **2290** includes one or more information storage devices **2291**, one or more computers **2292**, one or more networks **2295** and instructions **2296**. As to the one or more computers **2292**, each computer may include one or more processors (e.g., or processing cores) **2293** and memory **2294** for storing the instructions **2296**, for example, executable by at least one of the one or more processors. As an example, a computer may include one or more network interfaces (e.g., wired or wireless), one or more graphics cards, a display interface (e.g., wired or wireless), etc. The system **2290** may be specially configured to perform one or more portions of the method **2200** of FIG. **20**.

[0165] As an example, a computational framework may include a solver, which may be implemented via executable instructions. For example, consider a computational framework that includes a processor and memory accessible to the processor where executable instructions may be stored in the memory and accessed for execution by the processor to cause the computational framework to perform one or more actions. Such a computational framework may include one or more interfaces for receipt of information and/or for output of information, which may include values of parameters, an instruction, etc. As an example, a computational framework may be part of a controller. As an example, a computational framework may be part of a system.

[0166] As an example, various systems, methods, etc., may implement one or more ML models. As to types of ML models, consider one or more of a support vector machine (SVM) model, a k-nearest neighbors (KNN) model, an ensemble classifier model, a neural network (NN) model, etc. As an example, a machine learning model may be a deep learning model (e.g., deep Boltzmann machine, deep belief network, convolutional neural network, stacked auto-encoder, etc.), an ensemble model (e.g., random forest, gradient boosting machine, bootstrapped aggregation, AdaBoost, stacked generalization, gradient boosted regression tree, etc.), a neural network model (e.g., radial basis function network, perceptron, back-propagation, Hopfield network, etc.), a regularization model (e.g., ridge regression, least absolute shrinkage and selection operator, elastic net, least angle regression), a rule system model (e.g., cubist, one rule, zero rule, repeated incremental pruning to produce error reduction), a regression model (e.g., linear regression, ordinary least squares regression, stepwise regression, multivariate adaptive regression splines, locally estimated scatterplot smoothing, logistic regression, etc.), a Bayesian model (e.g., naïve Bayes, average on-dependence estimators, Bayesian belief network, Gaussian naïve Bayes, multinomial naïve Bayes, Bayesian network), a decision tree model (e.g., classification and regression tree, iterative dichotomiser 3, C4.5, C5.0, chi-squared automatic interaction detection, decision stump, conditional decision tree, M5), a dimensionality reduction model (e.g., principal component analysis, partial least squares regression, Sammon mapping, multidimensional scaling, projection pursuit, principal component regression, partial least squares discriminant analysis, mixture discriminant analysis, quadratic discriminant analysis, regularized discriminant analysis, flexible discriminant analysis, linear discriminant analysis, etc.), an instance model (e.g., k-nearest neighbor, learning vector quantization, self-organizing map, locally weighted learning, etc.), a clustering model (e.g., k-means, k-medians, expectation maximization, hierarchical clustering, etc.), etc.

[0167] As an example, a system may utilize one or more recurrent neural networks (RNNs). One type of RNN is referred to as long short-term memory (LSTM), which may be a unit or component (e.g., of one or more units) that may be in a layer or layers. A LSTM component may be a type of artificial neural network (ANN) designed to recognize patterns in sequences of data, such as time series data. When provided with time series data, LSTMs take time and sequence into account such that an LSTM may include a temporal dimension. For example, consider utilization of one or more RNNs for processing temporal data from one or more sources, optionally in combination with spatial data. Such an approach may recognize temporal patterns, which may be utilized for making predictions (e.g., as to a pattern or patterns for future times, etc.).

[0168] As an example, the TENSORFLOW framework (Google LLC, Mountain View, California) may be implemented, which is an open-source software library for dataflow programming that includes a symbolic math library, which may be implemented for machine learning applications that may include neural networks. As an example, the CAFFE framework may be implemented, which is a DL framework developed by Berkeley AI Research (BAIR) (University of California, Berkeley, California). As another example, consider the SCIKIT platform (e.g., scikit-learn), which utilizes the PYTHON programming language. As an example, a framework such as the APOLLO AI framework may be utilized (APOLLO.AI GmbH, Germany). As mentioned, a framework such as the PYTORCH framework may be utilized.

[0169] As an example, a training method may include various actions that may operate on a dataset to train a ML model. As an example, a dataset may be split into training data and test data where test data may provide for evaluation. A method may include cross-validation of parameters and best parameters, which may be provided for model training.

[0170] The TENSORFLOW framework may run on multiple CPUs and GPUs (with optional CUDA (NVIDIA Corp., Santa Clara, California) and SYCL (The Khronos Group Inc., Beaverton, Oregon) extensions for general-purpose computing on graphics processing units (GPUs)). TENSORFLOW is available on 64-bit LINUX, MACOS (Apple Inc., Cupertino, California), WINDOWS (Microsoft Corp., Redmond, Washington), and mobile computing platforms including ANDROID (Google LLC, Mountain View, California) and IOS (Apple Inc.) operating system-based platforms.

[0171] TENSORFLOW computations may be expressed as stateful dataflow graphs; noting that the name TENSORFLOW derives from the operations that such neural networks perform on multidimensional data arrays. Such arrays may be referred to as "tensors".

[0172] As an example, a method may include receiving data for drilling of a borehole in a subsurface environment according to a borehole trajectory for a target well at a field location in a basin; accessing offset well data for wells in the basin using at least a portion of the data, where the offset well data include identified formation tops with respect to well depth for a number of formations within the basin and mechanical specific energy drilling data with respect to well depth; aligning the offset well data using the formation tops to generate aligned offset well data that specify one or more drilling zones based on the mechanical specific energy drilling data; and performing the drilling based at least in part on the aligned offset well data.

[0173] As an example, mechanical specific energy (MSE) may be provided in one or more forms and/or may be utilized to derive one or more metrics; noting that a metric such as a bit index may be related to MSE.

[0174] As an example, a method may include performing drilling in a manner that includes setting one or more drilling parameters based at least in part on depth of one of one or more drilling zones.

[0175] As an example, one or more drilling zones may include one or more of an interbed formation and a hard stringer formation. As an example, one or more drilling zones may include a high shock and vibration interval.

[0176] As an example, a method may include determining one or more drilling zones by at least in part implementing a decision tree model.

[0177] As an example, a method may include determining a drilling risk for at least one of one or more drilling zones.

[0178] As an example, a method may include receiving offset well data and aligning offset well data where aligned offset well data may include uniaxial compressive strength (UCS) data. UCS data may provide a measure of a material's strength where, for example, UCS may be the maximum axial compressive stress that a right-cylindrical sample of material may withstand before failing. UCS may also be known as unconfined compressive strength of a material because confining stress as may be set to zero.

[0179] As an example, aligned offset well data may include abrasion data and/or impact data. As an example, a method may include generating simulated data for a target well using at least a portion of aligned offset well data.

[0180] As an example, a method may include accessing offset well data for wells in a basin in a manner that occurs automatically responsive to the receiving data for drilling (e.g., of a target well). In such an example, the accessing may include selecting the offset wells based at least in part on the location and/or selecting the offset wells based at least in part on a borehole trajectory (e.g., of a target well). As an example, a borehole trajectory may include a dogleg. As an example, drilling may include directional drilling, which may include drilling a borehole with a dogleg.

[0181] As an example, a method may include performing drilling in a manner that includes estimating real-time drill bit damage of a drill bit of a drillstring disposed in the borehole during performance of the drilling based at least in part on aligned offset well data and adjusting one or more drilling parameters of the drilling based on the estimating. For example, consider a method that may include determining a trip out time for tripping out the drillstring for servicing of the drill bit (e.g., a time or time range for pulling out of hole (POOH)). In such an example, a method may include calling for re-planning of operations due at least in part to tripping out prior to an originally planned tripping out time.

[0182] As an example, a method may include performing drilling in a manner that includes estimating real-time risk of an event occurring during performance of the drilling based at least in part on aligned offset well data and adjusting one or more drilling parameters of the drilling based on the estimating. In such an example, an event may be a likelihood of not meeting a planned metric. For example, consider a planned metric as to a dogleg where, if at a certain depth or range of depths, a planned dogleg is not met (e.g., as may be characterized by dogleg severity, etc.), a method may call for adjusting one or more drilling parameters and, for example, calling for re-planning of drilling.

[0183] As an example, a method may include performing drilling in a manner that includes implementing a controller that includes levels of automation. In such an example, a framework may provide for outputting recommendations and/or control instructions that may call for adjusting a level of automation. For example, if a hazard is approach that may demand human expertise to handle, then a level of automation may be changed to involve manual control; whereas, if planned metrics and real-time metrics may be substantially in agreement, a framework may recommend or issue an instruction to change a level of automation to be more automated (e.g., less human involvement, etc.).

[0184] As an example, a framework may provide for output of one or more alert tracks, which may include, for example, a level of automation track that may provide for recommendations and/or issuance of instructions for changing a level of automation. In such an approach, the level of automation track may be viewed in the context of various other tracks such that an overall assessment may be made visually by an operator. For example, if a framework does not indicate a change to a level of automation for a depth involving a particular challenging type of formation to drill, then an operator may call for implementing a level of automation until a time (e.g., a depth) at which the operator may manually intervene. In such an example, a GUI may be interactive and allow a driller to tailor (e.g., customize) one or more levels of automation for one or more depths (e.g., ranges of depths) of drilling of a borehole or a portion thereof.

[0185] As an example, a system may include one or more processors; memory accessible to at least one of the one or more processors; processor-executable instructions stored in the memory and executable to instruct the system to: receive data for drilling of a borehole in a subsurface environment according to a borehole trajectory for a target well at a field location in a basin; access offset well data for wells in the basin using at least a portion of the data, where the offset well data include identified formation tops with respect to well depth for a number of formations within the basin and mechanical specific energy drilling data with respect to well depth; align the offset well data using the formation tops to generate aligned offset well data that specify one or more drilling zones based on the mechanical specific energy drilling data; and perform the drilling based at least in part on the aligned offset well data.

[0186] As an example, one or more computer-readable storage media may include processor-executable instructions to instruct a computing system to: receive data for drilling of a borehole in a subsurface environment according to a borehole trajectory for a target well at a field location in a basin; access offset well data for wells in the basin using at least a portion of the data, where the offset well data include identified formation tops with respect to well depth for a number of formations within the basin and mechanical specific energy drilling data with respect to well depth; align the offset well data using the formation tops to generate aligned offset well data that specify one or more drilling zones based on the mechanical specific energy drilling data; and perform the drilling based at least in part on the aligned offset well data.

[0187] As an example, a computer program product that may include computer-executable instructions to instruct a computing system to perform one or more methods such as one or more of the methods described herein (e.g., in part, in whole and/or in various combinations).

[0188] In some embodiments, a method or methods may be executed by a computing system. FIG. **23** shows an example of a system **2300** that may include one or more computing systems **2301-1**, **2301-2**, **2301-3** and **2301-4**, which may be operatively coupled via one or more networks **2309**, which may include wired and/or wireless networks. As shown, the system **2300** may include one or more other components **2308**.

[0189] As an example, a system may include an individual computer system or an arrangement of distributed computer systems. In the example of FIG. **23**, the computer system **2301-1** may include one or more modules **2302**, which may be or include processor-executable instructions, for example, executable to perform various tasks (e.g., receiving information, requesting information, processing information, simulation, outputting information, etc.).

[0190] As an example, a module may be executed independently, or in coordination with, one or more processors **2304**, which is (or are) operatively coupled to one or more storage media **2306** (e.g., via wire, wirelessly, etc.). As an example, one or more of the one or more processors **2304** may be operatively coupled to at least one of one or more network interface **2307**. In such an example, the computer system **2301-1** may transmit and/or receive information, for example, via the one or more networks **2309** (e.g., consider one or more of the Internet, a private network, a cellular network, a satellite network, etc.). As shown, one or more other components **2308** may be included in the computer system **2301-1**.

[0191] As an example, the computer system **2301-1** may receive from and/or transmit information to one or more other devices, which may be or include, for example, one or more of the computer systems **2301-2**, etc. A device may be located in a physical location that differs from that of the computer system **2301-1**. As an example, a location may be, for example, a processing facility location, a data center location (e.g., server farm, etc.), a rig location, a wellsite location, a downhole location, etc.

[0192] As an example, a processor may be or include a microprocessor, microcontroller, processor module or subsystem, programmable integrated circuit, programmable gate array, or another control or computing device.

[0193] As an example, the storage media **2306** may be implemented as one or more computer-readable or machine-readable storage media. As an example, storage may be distributed within and/or across multiple internal and/or external enclosures of a computing system and/or additional computing systems.

[0194] As an example, a storage medium or storage media may include one or more different forms of memory including semiconductor memory devices such as dynamic or static random access memories (DRAMs or SRAMs), erasable and programmable read-only memories (EPROMs), electrically erasable and programmable read-only memories (EEPROMs) and flash memories, magnetic disks such as fixed, floppy and removable disks, other magnetic media including tape, optical media such as compact disks (CDs) or digital video disks (DVDs), BLUERAY disks, or other types of optical storage, or other types of storage devices.

[0195] As an example, a storage medium or media may be located in a machine running machine-readable instructions, or located at a remote site from which machine-readable instructions may be downloaded over a network for execution.

[0196] As an example, various components of a system such as, for example, a computer system, may be implemented in hardware, software, or a combination of both hardware and software (e.g., including firmware), including one or more signal processing and/or application specific integrated circuits.

[0197] As an example, a system may include a processing apparatus that may be or include a general-purpose processors or application specific chips (e.g., or chipsets), such as ASICs, FPGAs, PLDs, or other appropriate devices.

[0198] As an example, a device may be a mobile device that includes one or more network interfaces for communication of information. For example, a mobile device may include a wireless network interface (e.g., operable via IEEE 802.11, ETSI GSM, BLUETOOTH, satellite, etc.). As an example, a mobile device may include components such as a main processor, memory, a display, display graphics circuitry (e.g., optionally including touch and gesture circuitry), a SIM slot, audio/video circuitry, motion processing circuitry (e.g., accelerometer, gyroscope), wireless LAN circuitry, smart card circuitry, transmitter circuitry, GPS circuitry, and a battery. As an example, a mobile device may be configured as a cell phone, a tablet, etc. As an example, a method may be implemented (e.g., wholly or in part) using a mobile device. As an example, a system may include one or more mobile devices.

[0199] As an example, a system may be a distributed environment, for example, a so-called "cloud" environment where various devices, components, etc. interact for purposes of data storage, communications, computing, etc. As an example, a device or a system may include one or more components for communication of information via one or more of the Internet (e.g., where communication occurs via one or more Internet protocols), a cellular network, a satellite network, etc. As an example, a method may be implemented in a distributed environment (e.g., wholly or in part as a cloud-based service).

[0200] As an example, information may be input from a display (e.g., consider a touchscreen), output to a display or both. As an example, information may be output to a projector, a laser device, a printer, etc. such that the information may be viewed. As an example, information may be output stereographically or holographically. As to a printer, consider a 2D or a 3D printer. As an example, a 3D printer may include one or more substances that may be output to construct a 3D object. For example, data may be provided to a 3D printer to construct a 3D representation of a subterranean formation. As an example, layers may be constructed in 3D (e.g., horizons, etc.), geobodies constructed in 3D, etc. As an example, holes, fractures, etc., may be constructed in 3D (e.g., as positive structures, as negative structures, etc.).

[0201] Although only a few examples have been described in detail above, those skilled in the art will readily appreciate that many modifications are possible in the examples. Accordingly, all such modifications are intended to be included within the scope of this disclosure as defined in the following claims. In the claims, means-plus-function clauses are intended to cover the structures described herein as performing the recited function and not only structural equivalents, but also equivalent structures. Thus, although a nail and a screw may not be structural equivalents in that a nail employs a cylindrical surface to secure wooden parts together, whereas a screw employs a helical surface, in the environment of fastening wooden parts, a nail and a screw may be equivalent structures.

What is claimed is:

1. A method comprising:

receiving data for drilling of a borehole in a subsurface environment according to a borehole trajectory for a target well at a field location in a basin;

accessing offset well data for wells in the basin using at least a portion of the data, wherein the offset well data comprise identified formation tops with respect to well depth for a number of formations within the basin and mechanical specific energy drilling data with respect to well depth;

aligning the offset well data using the formation tops to generate aligned offset well data that specify one or more drilling zones based on the mechanical specific energy drilling data; and

performing the drilling based at least in part on the aligned offset well data.

2. The method of claim **1**, wherein performing the drilling comprises setting one or more drilling parameters based at least in part on depth of one of the one or more drilling zones.

3. The method of claim **1**, wherein the one or more drilling zones comprise one or more of an interbed formation and a hard stringer formation.

4. The method of claim **1**, wherein the one or more drilling zones comprise a high shock and vibration interval.

5. The method of claim **1**, comprising determining the one or more drilling zones by at least in part implementing a decision tree model.

6. The method of claim **1**, comprising determining a drilling risk for at least one of the one or more drilling zones.

7. The method of claim **1**, wherein the aligned offset well data comprise uniaxial compressive strength (UCS) data.

8. The method of claim **1**, wherein the aligned offset well data comprise abrasion data.

9. The method of claim **1**, wherein the aligned offset well data comprise impact data.

10. The method of claim **1**, comprising generating simulated data for the target well using at least a portion of the aligned offset well data.

11. The method of claim **1**, wherein the accessing occurs automatically responsive to the receiving data for drilling.

12. The method of claim **11**, wherein the accessing comprises selecting the offset wells based at least in part on the location.

13. The method of claim **11**, wherein the accessing comprises selecting the offset wells based at least in part on the borehole trajectory.

14. The method of claim **13**, wherein the borehole trajectory comprises a dogleg.

15. The method of claim **1**, wherein the performing comprises estimating real-time drill bit damage of a drill bit of a drillstring disposed in the borehole during performance of the drilling based at least in part on the aligned offset well data and adjusting one or more drilling parameters of the drilling based on the estimating.

16. The method of claim **15**, comprising determining a trip out time for tripping out the drillstring for servicing of the drill bit.

17. The method of claim **1**, wherein the performing comprises estimating real-time risk of an event occurring during performance of the drilling based at least in part on the aligned offset well data and adjusting one or more drilling parameters of the drilling based on the estimating.

18. The method of claim **1**, wherein the performing comprises implementing a controller that comprises levels of automation.

19. A system comprising:

one or more processors;

memory accessible to at least one of the one or more processors;

processor-executable instructions stored in the memory and executable to instruct the system to:

receive data for drilling of a borehole in a subsurface environment according to a borehole trajectory for a target well at a field location in a basin;

access offset well data for wells in the basin using at least a portion of the data, wherein the offset well data comprise identified formation tops with respect to well depth for a number of formations within the basin and mechanical specific energy drilling data with respect to well depth;

align the offset well data using the formation tops to generate aligned offset well data that specify one or more drilling zones based on the mechanical specific energy drilling data; and

perform the drilling based at least in part on the aligned offset well data.

20. One or more computer-readable storage media comprising processor-executable instructions to instruct a computing system to:

receive data for drilling of a borehole in a subsurface environment according to a borehole trajectory for a target well at a field location in a basin;

access offset well data for wells in the basin using at least a portion of the data, wherein the offset well data comprise identified formation tops with respect to well depth for a number of formations within the basin and mechanical specific energy drilling data with respect to well depth;

align the offset well data using the formation tops to generate aligned offset well data that specify one or more drilling zones based on the mechanical specific energy drilling data; and

perform the drilling based at least in part on the aligned offset well data.

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